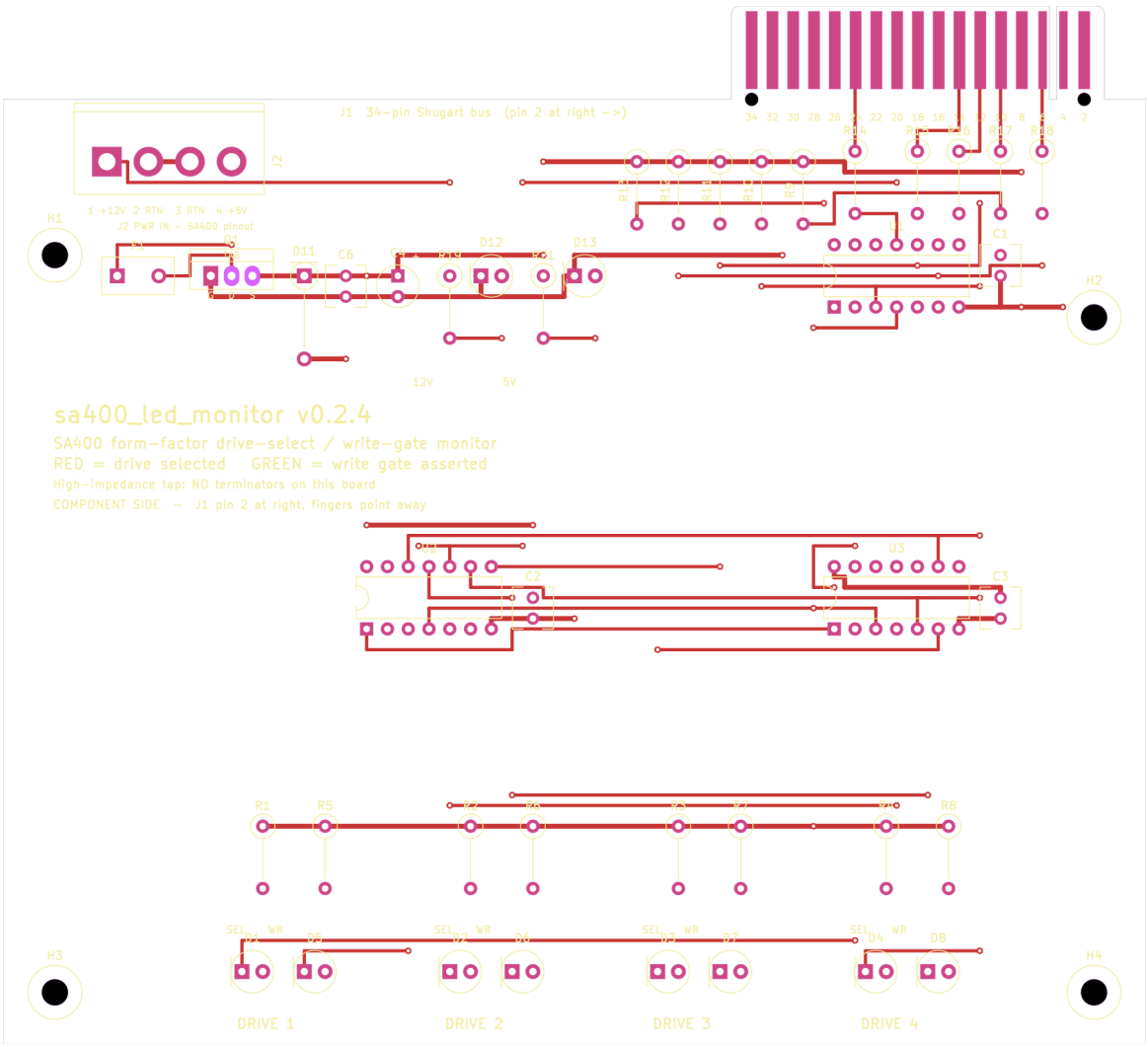
sa400_led_monitor

SA400 form-factor drive-select / write-gate LED monitor

Complete design document — v0.2.4

Board	5.5 × 5.0 in, 2 layers, Shugart 34-pin card edge, AMP Mate-N-Lok power
Logic	74LS14 receivers, 74LS06 (red, drive selected), 74LS00 (green, write gate)
Power	SA400 usage: +5 V logic, +12 V present; PTC, P-MOSFET reverse switch, 5.6 V clamp
Target	TRS-80 Model I Expansion Interface floppy bus, JLCPCB fabrication
Owner	Ven
Created	2026-09-05
Updated	2026-10-19
Status	KiCad 10.0.6 DRC 0 / ERC 0; three physical measurements outstanding (Section 9)



Generated from the Python design source in `gen/`. Every dimension in this document is a value in `gen/config.py`; items marked **ASSUMED** are design decisions not yet confirmed against hardware.

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1 Overview and specification

1.1 What the board does

The board plugs onto the 34-conductor Shugart floppy bus exactly as a fifth drive would and shows, on eight LEDs, which of the four drives the host is addressing and whether it is writing:

LED	Colour	Lights when
D1–D4 “SEL”	red	the host asserts DRIVE SELECT 1–4
D5–D8 “WR”	green	the host asserts DRIVE SELECT n **and** WRITE GATE

Per drive n, with the bus-side signals active-low and the post-receiver signals active-high:

$$\mathrm{RED}_n = \mathrm{DS}_n \qquad \mathrm{GREEN}_n = \mathrm{DS}_n \cdot \mathrm{WG}$$

Red stays on during a write; green joins it. Two yellow LEDs beside the power header show that +12 V and +5 V are present.

1.2 Specification

Item	Value
Bus interface	Shugart 34-pin card edge (J1), high-impedance tap, no terminators, never drives the bus
Signals monitored	DRIVE SELECT 1–4 (pins 10, 12, 14, 6), WRITE GATE (pin 24)
Power	+5 V from J2 pin 4 (Commercial Mate-N-Lok 4-pos, SA400 pinout, component side since 0.2.4) → PTC → P-MOSFET → logic, ≈ 45 mA; +12 V pin lights an indicator (4.5 mA)
Protection	PTC 0.5 A, IRF4905 P-MOSFET reverse-polarity switch (millivolts of drop), 5.6 V Zener clamp on +5 V, 470 Ω series in each bus line
Indicators	8 × 5 mm LEDs, ≈ 5.5 mA each with 470 Ω (1.5 kΩ for high-efficiency LEDs)
Logic	74LS14, 74LS00, 74LS06 — all DIP-14, socketed
Board	139.70 × 127.00 mm, 1.6 mm, 2 layers, 1 oz, ENIG + hard-gold bevelled fingers
Mounting	4 × M3 (deviation from the SA400’s snap clamps)
Fab	JLCPCB, gerbers in `output/gerbers/`

1.3 Package contents

Path	Contents
`kicad/`	KiCad 10 project: `.kicad_pro`, `.kicad_sch`, `.kicad_pcb`, `sa400.pretty/`, `sa400.kicad_sym`, lib tables
`output/gerbers/`	Gerber X2 + Excellon plotted by KiCad 10.0.6
`output/drc.json, erc.json`	KiCad 10 DRC (with schematic parity) and ERC reports, both zero
`output/*.step, *.pdf, *.png`	3D STEP, schematic print, board renders
`docs/`	Markdown originals of every section in this document
`gen/`	Python generator: `config.py` (all dimensions), `design.py` (placement, netlist), `router.py`, writers, `postprocess.py`, `check.ps1`

2 Theory of operation

2.1 The Shugart bus from first principles

Physical layer. The interface defined in the Shugart SA400L OEM manual (P/N 39019-1, section 3) is a 34-conductor ribbon cable that lands on a card-edge connector, J1, on each drive. The 34 contacts are 17 pairs: even numbers 2–34 on the component side, odd numbers 1–33 directly beneath them on the solder side. **Every odd contact is ground.** Signals therefore alternate with grounds along the ribbon, which keeps crosstalk low on a cable that may run several feet. The five contacts this board uses:

J1 pin	Signal	Direction
10	DRIVE SELECT 1	host → drive
12	DRIVE SELECT 2	host → drive
14	DRIVE SELECT 3	host → drive
6	DRIVE SELECT 4	host → drive
24	WRITE GATE	host → drive

All other even contacts (INDEX, STEP, DIRECTION, MOTOR ON, WRITE DATA, READ DATA, TRACK 00, WRITE PROTECT and the spares) exist as fingers so the connector seats normally, but have no copper path onto the board.

Open-collector. The manual specifies a 7438 as the host's driver. A 7438 output contains only a transistor to ground. To assert a line it turns that transistor on and pulls the line to about 0.4 V; to release it, it turns the transistor off and the line floats. Something else must pull the line to +5 V — the terminator.

Terminator. The *last* drive on the cable carries a 150 Ω pull-up pack from every signal line to +5 V (SA400L §3.2.1). That fixes the current the host driver sinks when it asserts a line:

$$I = \frac{5\,\mathrm{V} - 0.4\,\mathrm{V}}{150\,\Omega} \approx 31\ \mathrm{mA}$$

A 7438 is rated to sink 48 mA, so one terminator is fine and two would be marginal. **This is why the board carries no terminators**: a second 150 Ω pack in parallel would halve the resistance and push the host driver past its rating.

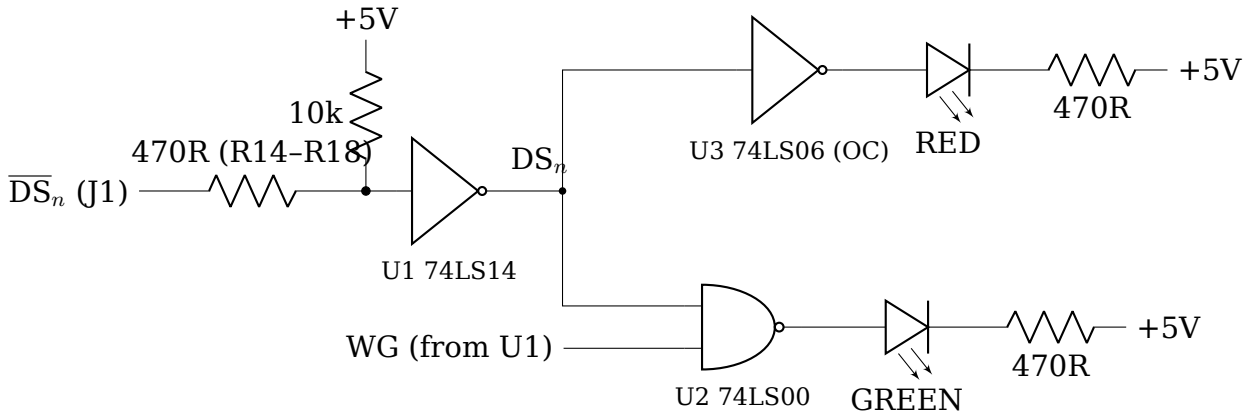
Active-low. Because pulling down is the only strong action a driver can take, "asserted" is the low state. Bus-side nets are written with an overline, e.g. $\overline{\mathrm{DS1}}$, $\overline{\mathrm{WG}}$.

Line voltage	Meaning
≈ 0.4 V (host pulling down)	signal **asserted**
≈ 5 V (terminator pulling up)	signal released

Receiver. The manual recommends a 7414 hex Schmitt-trigger inverter. It inverts, so the logic becomes active-high after it; and it has hysteresis — it switches high at about 1.7 V and low at about 0.9 V, and the 0.8 V gap swallows ringing and ground bounce on a long ribbon. That is exactly why Shugart specifies it.

Timing. The host asserts one DRIVE SELECT for as long as it talks to that drive: at minimum a few hundred microseconds for a step, typically tens to hundreds of milliseconds for a read or write. WRITE GATE is asserted only while write current flows, and only with a DRIVE SELECT already active (the manual requires 100 μs of lead). The LEDs follow the lines directly with no stretching — what you see is the truth. If very short accesses prove too brief to notice, a pulse-stretcher is a listed v0.3 option.

2.2 Signal path on the board



Series resistors — R14–R18, 470 Ω. Each monitored finger reaches the rest of the board only through a 470 Ω resistor (nets $\overline{\text{DS}}_n$ on the finger side, DS$_{n,\text{in}}$ on the receiver side). Their purpose is protection of the host, analysed in Section 2.3; their cost is 0.19 V of drop at the LS14's 0.4 mA input current, which leaves a 0.4 V bus low at 0.59 V — well under the Schmitt lower threshold.

Correction, 2026-10-19 design review. The 0.59 V counts only the receiver's input current. R9–R13 sit on the *receiver* side of R14–R18, so when the bus is low their ≈ 0.44 mA also leaves through the series resistor: worst case $0.4 + (0.4 + 0.44)\,\text{mA} \times 470\,\Omega \approx$ **0.79 V** at the LS14 input, against SN74LS14 V_{T-} = 0.5 V min / 0.8 V typ / 1.0 V max. Typical parts ($V_{OL} \approx 0.25$ V, $I_{IL} \approx 0.2$ mA) give ≈ 0.55 V, so the board works on *typical* margin, not worst-case margin. It is measured on the real bus in TEST_PLAN.md step 3.3.7. Remedies: a 74HCT14 in U1 removes the I_{IL} term with no board change (its input current is μA); moving the pull-ups to the finger side of the series resistor removes the pull-up term (planned for v0.3). With both, the receiver sees the bus voltage itself.

Receiver stage — U1, 74LS14. Each bus line goes from its series resistor to one input of U1. The output is the inverted, cleaned-up signal: DS$_n$ is high when drive n is selected, WG is high when the host is writing.

Loading: a 74LS14 input sources at most 0.4 mA when held low. Against the 31 mA the terminator already makes the host sink, that is 1.3 % — the board is invisible to the bus.

Pull-ups: R9–R13, 10 kΩ to +5 V on each input. On a terminated bus they do almost nothing (10 kΩ ∥ 150 Ω = 148 Ω). They define the inputs when the board is powered with no cable, or on a cable whose last drive has had its terminator pulled; without them the inputs float and the LEDs flicker. They add 0.5 mA to what the host sinks.

Spare gate: gate 6 of U1 has its input (pin 13) tied to ground and its output (pin 12) open. Floating TTL inputs usually read high but can oscillate and add supply noise.

Red — U3, 74LS06 open-collector inverter. RED_n must light whenever DS_n is high. When DS_n goes high the 74LS06 output pulls the LED cathode to ground and the LED conducts through 470 Ω from +5 V; otherwise the output transistor is off, the cathode floats up, no current flows. A 74LS06 was chosen over a spare 74LS14 gate because it sinks 40 mA (74LS14: 8 mA), leaving headroom for brighter LEDs, and because an open-collector output has nothing fighting the LED chain.

Green — U2, 74LS00 quad NAND. GREEN_n must light only when drive n is selected **and** WRITE GATE is active. A NAND output is low exactly when both inputs are high — precisely the condition, since both DS_n and WG are active-high after U1. The low output sinks the green LED the same way.

DS_n	WG	RED (U3 out)	GREEN (U2 out)	What you see
0	x	high (off)	high (off)	nothing — drive not selected
1	0	**low (on)**	high (off)	red — selected, reading or seeking
1	1	**low (on)**	**low (on)**	red **and** green — writing

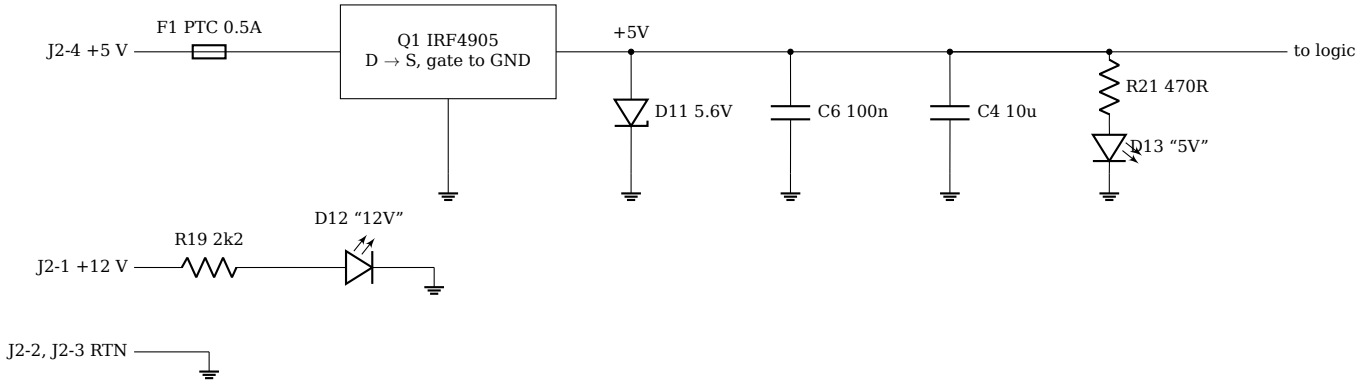
LED current. The board is built with the standard-brightness 5 mm LEDs JLCPCB stocks (Kingbright L-53ID red, WP7113GD green, WP483YDT yellow), which want 5–10 mA, so R1–R8 and R21 are 470 Ω. With the output saturated at about 0.4 V and a red LED dropping ≈ 1.9 V:

$$I = \frac{5.0 - 0.4 - 1.9}{470} \approx 5.7 \text{ mA}$$

Green parts drop ≈ 2.1 V, giving ≈ 5.3 mA — inside the 74LS00's 8 mA sink ($V_{OL} \leq 0.5$ V). With high-efficiency LEDs (Kingbright WP7113LID / WP7113LGD, bright at 2 mA) use 1.5 kΩ instead (≈ 1.8 mA).

2.3 Power and protection

J2 carries the SA400 pinout, and the board uses the two rails exactly as the drive does: **+5 V for the logic**, **+12 V present** on the other pair. The SA400 spends its +12 V on motors; this board has none, so +12 V lights an indicator and nothing else.



Part	Job	Why this value
F1 PTC 0.5 A hold / 1 A trip	limits energy into any downstream fault; resets when it clears	board draws ≈ 45 mA
Q1 IRF4905PBF P-MOSFET (TO-220, G-D-S), drain in / source out / gate to GND	reverse-polarity switch: correct polarity turns the channel on ($V_{GS} = -5$ V, **millivolts of drop** at 45 mA); reversed, the body diode is blocked and V_{GS} is positive, nothing conducts	a series diode would drop 0.35–0.7 V, below the 74LS minimum V_{CC} of 4.75 V. IRF4905 is JLCPCB stock ($V_{GS(th)}$ -2 to -4 V, 20 mΩ at -10 V; only partly enhanced at -5 V, irrelevant at 45 mA); onsemi NDP6020P (logic-level, 50 mΩ at -4.5 V) is the drop-in alternative. Metal tab toward the J2 edge; gate = square pad
D11 1N5339B 5.6 V 5 W Zener	clamps the rail if the wrong voltage arrives (e.g. +12 V on the +5 V pin) and pulls current until F1 trips	74LS absolute maximum V_{CC} is 7 V
C6 100 nF, C4 10 μF	rail decoupling at the entry point	—
R19 2.2 kΩ + D12	"+12 V present" indicator, ≈ 4.5 mA	confirms a real drive cable with both rails
R21 470 Ω + D13	"+5 V rail up" indicator, ≈ 6 mA	shows the PTC has not tripped
R14–R18 470 Ω series	limits any current the board can push into the bus	worst case 11 mA per line, see below

Current budget. U1 10 mA, U2 4 mA, U3 10 mA, two drive LEDs 11 mA, pull-ups 2.5 mA, D13 6 mA: ≈ **45 mA** on +5 V (≈ 80 mA if all eight drive LEDs were lit); 4.5 mA on +12 V.

Why the series resistors protect the Expansion Interface. The EI drives the bus with 7438 open-collector gates rated 5.5 V and 48 mA sink. With 470 Ω between each finger and everything else on the board:

- a failed receiver can push at most $(5.6 - 0.4)/470 \approx 11$ mA into the EI's driver (the rail is held below 5.6 V by D11); with the terminator's 31 mA that is 42 mA, inside the 48 mA rating;
- hot-plug transients on the card edge meet 470 Ω before the LS14 input;
- the receiver still sees a clean low: 0.4 mA through 470 Ω is 0.19 V. Corrected 2026-10-19: with the pull-up current included the worst case is ≈ 0.79 V against a 0.5 V minimum threshold — see the correction under *Series resistors*; the fault-current bound above is unaffected.

Fault table.

Fault	What happens	Effect on the EI
+12 V / +5 V swapped in a miswired cable	D11 clamps at ≈ 6 V. Corrected 2026-10-19: the current is *not* ≈ 1 A — a cold 0.5 A PTC is ≈ 0.5 Ω, so until F1 trips it is whatever the supply, cable and Q1 will pass (amps). F1 trips well inside a second; D11 may fail, normally short, leaving the board dead with the rail at 0 V. D12 dark, D13 dim	≤ 11 mA per line until F1 trips; a shorted D11 afterwards means 0 mA
Reversed +5 V	Q1 stays off; nothing conducts	none
Reversed +12 V	only R19/D12 see it. Corrected 2026-10-19: D12 does not simply block — a 5 mm LED is rated ≈ 5 V reverse, so it is driven ≈ 7 V into reverse breakdown at ≈ 3 mA through R19. It will most likely survive; v0.3 adds a 1N4148 in series.	none
Spike on +5 V	D11 absorbs (5 W); F1 limits	none
LS14 input shorted to rail	470 Ω limits to ≈ 11 mA	within the 7438 rating
Board unpowered, EI on	≈ 0.6 mA per line into the dead inputs	0.1 V on a 150 Ω line
Board powered, EI off	≈ 0.5 mA per line into the EI	harmless

Ground is common to both PSU returns (J2 pins 2 and 3) and to the bus (fifteen odd fingers), as in a drive; the receivers judge the bus lines against that ground.

2.4 Why 74LS rather than 74HCT

74HCT would load the bus less still, but: the board is meant to look like a 1979 drive; 74HCT outputs are rated only 4 mA per pin against the 74LS06's 40 mA; and 74LS14 is the modern equivalent of the 7414 Shugart specifies. 74HCT14/00/06 are pin-compatible substitutes if the LED resistors are raised to keep each output under 4 mA.

2.5 What the board deliberately does not do

It does not terminate the bus, does not drive anything onto it (no IC output touches J1), draws only a 4.5 mA indicator from +12 V, and does not stretch LED pulses.

3 Mechanical

Origin is the top-left corner of the outline in KiCad; $+y$ runs toward the front (LED) edge. Every value is in `gen/config.py`; change it there and rebuild.

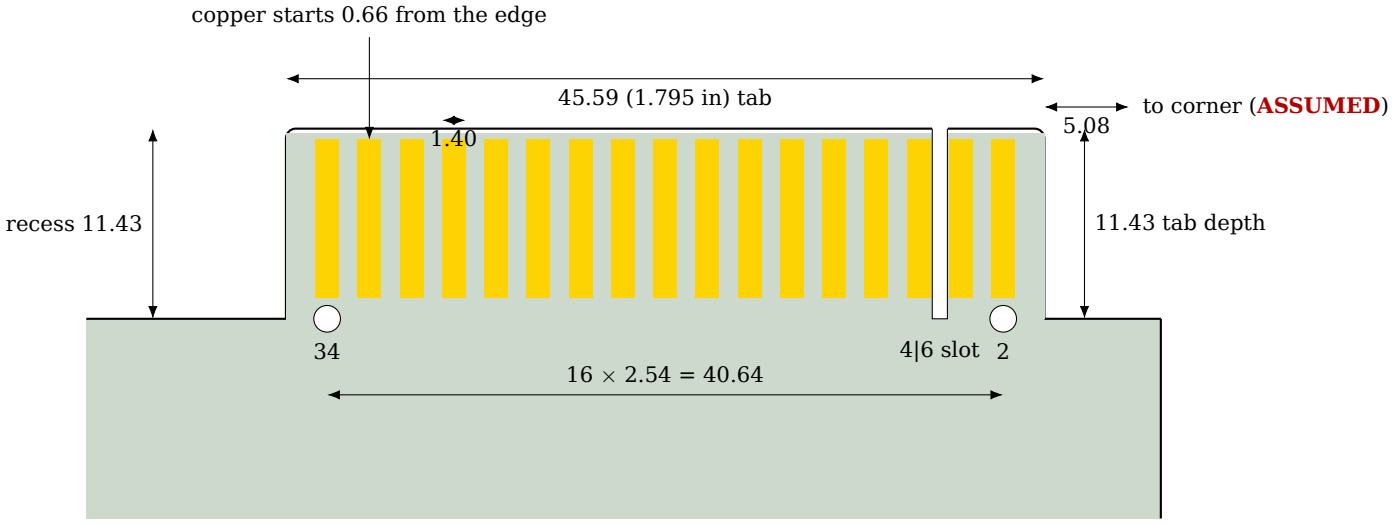
3.1 Board

Item	Value	Source
Outline	139.70 × 127.00 mm (5.5 × 5.0 in)	Ven
Thickness	1.6 mm	3M TS-0010 (1.37–1.78); SA400L fig 3-2 (0.062 in)
Rear edge	recessed 11.43 mm along its full length; only the J1 tab reaches $y = 0$	Ven's SA400 PCB photo

3.2 J1 card-edge tab

The tab follows 3M TS-0010-14 sheet 3, "Recommended PCB Layout" for the 3463 34-way connector — the connector Shugart names as P1; Shugart's figure 3-2 is a copy of it.

Item	Value	Source
Tab width (dim A)	45.59 mm (1.795 ± .005 in), x = 89.03–134.62	3M
Tab depth	11.43 mm (.450 min)	3M / SA400L
Tab corner radius	1.02 mm (R.040)	3M
Contact pitch	2.54 mm × 16, non-cumulative	3M
Contact width	1.40 mm (.055 ± .005)	3M
Contact reach	10.16 mm (.400 ± .015) from the edge	3M
Contact setback from edge	**0.66 mm** (3M .026); JLCPCB needs ≥ 0.6 mm for the 30° bevel	3M / JLCPCB
Pin 2 centre	x = 132.14 (2.475 from the tab edge)	fingers centred on A
Pin 34 centre	x = 91.50	derived
Handedness	pin 2 at the right viewed from the component side, fingers away; odd pins on the solder side	SA400L §3.2 + photo
Pin-2 side, tab edge to board corner	5.08 mm	**ASSUMED** ±1.5, scaled from photo
Key slot	0.91 mm wide × 11.43 deep between pins 4 and 6, x = 128.33	width 3M/TEAC; depth Ven
Pads 3–6	narrowed to 1.0 mm → 0.32 mm copper to slot	derived
Clip holes	2 × ⌀1.6 mm **non-plated** at (132.14, 11.43) and (91.50, 11.43): on the tab-root line, in line with pins 2 and 34, for the connector's retention clips	photo; ⌀ from fig 3-2 ".063 NOM (2X)"
Edge bevel	30°, 0.6 mm (JLCPCB); 3M draws 45° × 1.02	fab option

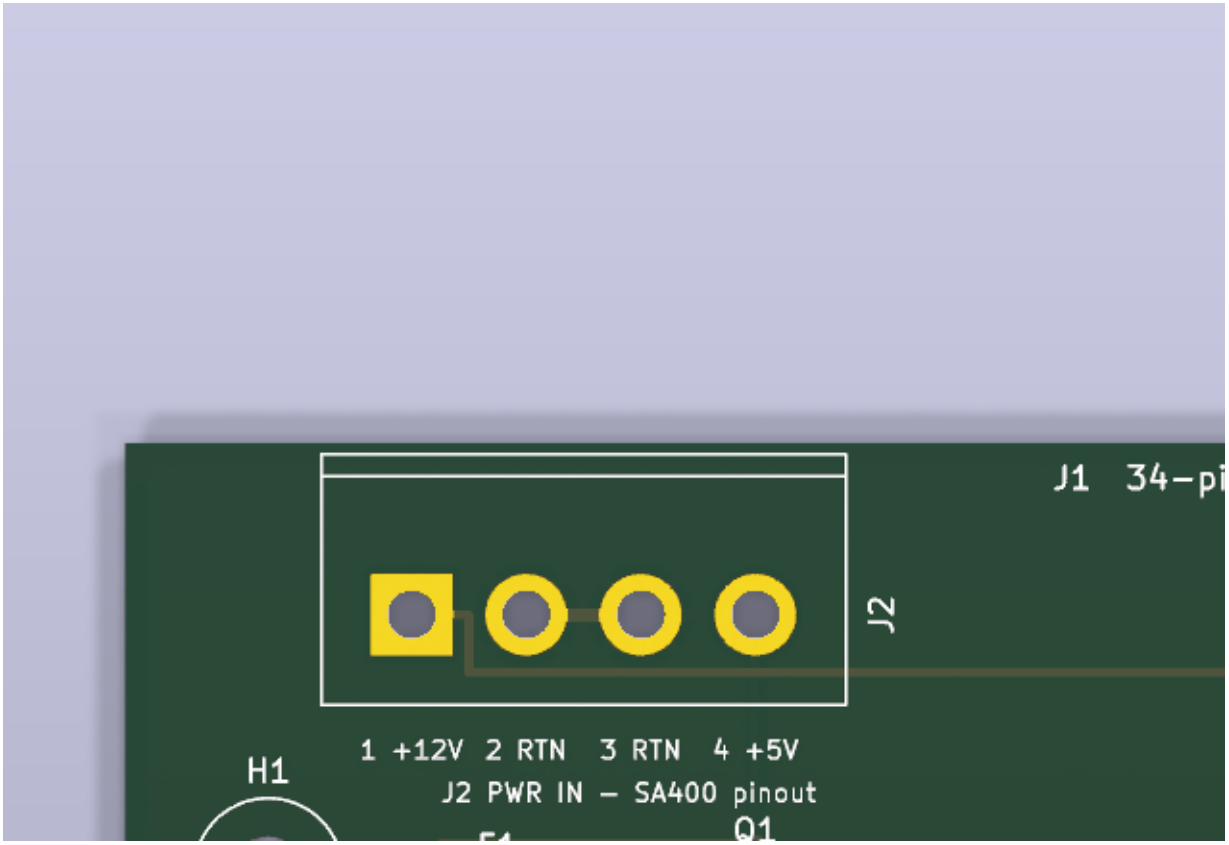


Component side, fingers pointing up (toward the rear of the drive). Even pins shown; odd pins are directly beneath on the solder side. Two ⌀1.6 NPTH clip holes on the root line. Dimensions in mm.

3.3 J2 power header

TE Commercial Mate-N-Lok 4-position *right-angle* header 1-641737-1 on the *component* side, mating face at the recessed rear edge (the vertical 350211-1 fits the same pads). The SA400 carries its header on the solder side and 0.2.0–0.2.3 copied that; JLCPCB's single-sided (Economic) assembly then left J2 — the only bottom-side part — unpopulated, so 0.2.4 moved it. Moving a header between sides *mirrors* it: pin 1 is now at the H1 (corner) end. Both layouts rest on the same **ASSUMED** header handedness, which is settled on the bench, not on paper: header in its holes, drive plug on, continuity from the plug's yellow (+12 V) wire to the pad marked +12V; if it lands on +5V, the header belongs on the other side (or `J2_PIN1_X`/`J2_PIN_DIR` flip and the board is rebuilt).

Item	Value	Source
Pin pitch	5.08 mm	TE
Pin row	y = 19.05 (7.62 behind the recessed edge)	**ASSUMED**
Pin 1	x = 12.70 (H1 end); pins toward the board centre, pin 4 at x = 27.94	**ASSUMED** — mirror of the 0.2.x solder-side layout; bench check above
Drill / pad	2.1 / 3.6 mm	TE lists the 350211-1 post at 1.6 mm [.063]; **ASSUMED** until ENG_CD_641737 is checked
Pinout	1 +12 V (indicator), 2 +12 RTN, 3 +5 RTN, 4 +5 V **(logic supply)** — SA400 pinout; pin 1 = square pad	SA400L §3.3
Housing	right-angle, component side; mating face at the recessed rear edge (y = 11.93), body to 4 mm behind the pin row (y = 23.05); outline on F.SilkS / F.Fab	TE 1-641737-1 / 350211-1
Labels	component side, below the housing: 1 +12V / 2 RTN / 3 RTN / 4 +5V and "J2 PWR IN – SA400 pinout"	v0.2.4

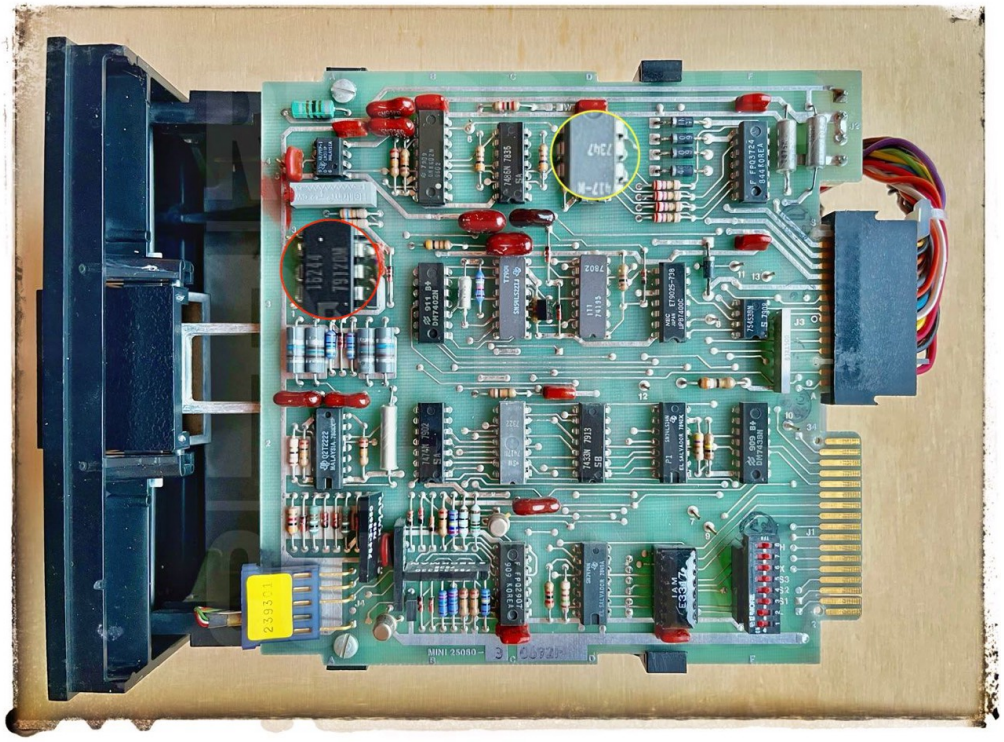


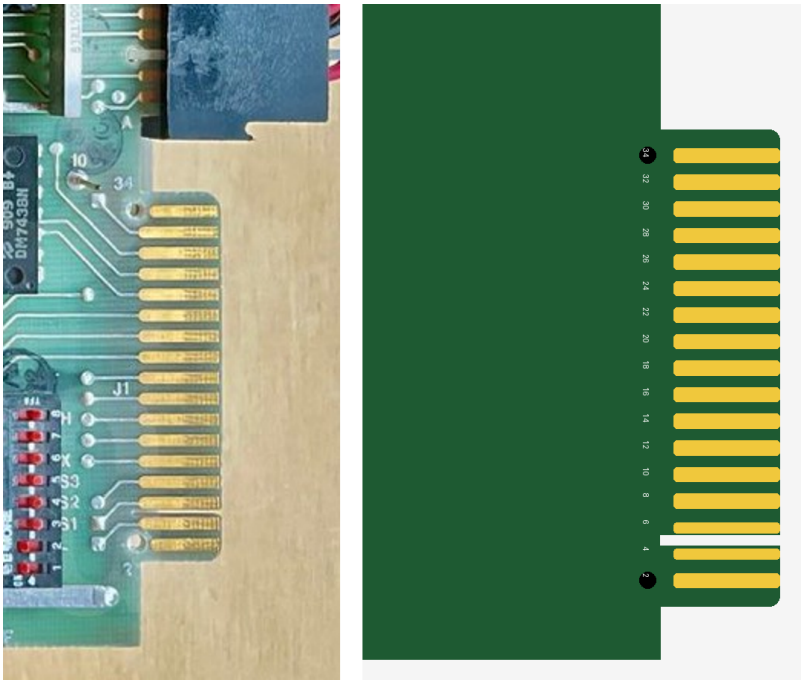
J2 as rendered on the component side: housing outline toward the rear edge, pin 1 (square) at the H1 end, labels below the housing. Power row at y = 33.02.

3.4 Mounting

H1–H4: ⌀3.2 mm NPTH at (6.35, 30.48), (133.35, 38.10), (6.35, 120.65), (133.35, 120.65). The real SA400 PCB is held by four snap clamps (P/N 54036); screw holes are a deliberate deviation for a standalone board. H2 is lowered to clear the cable connector body.

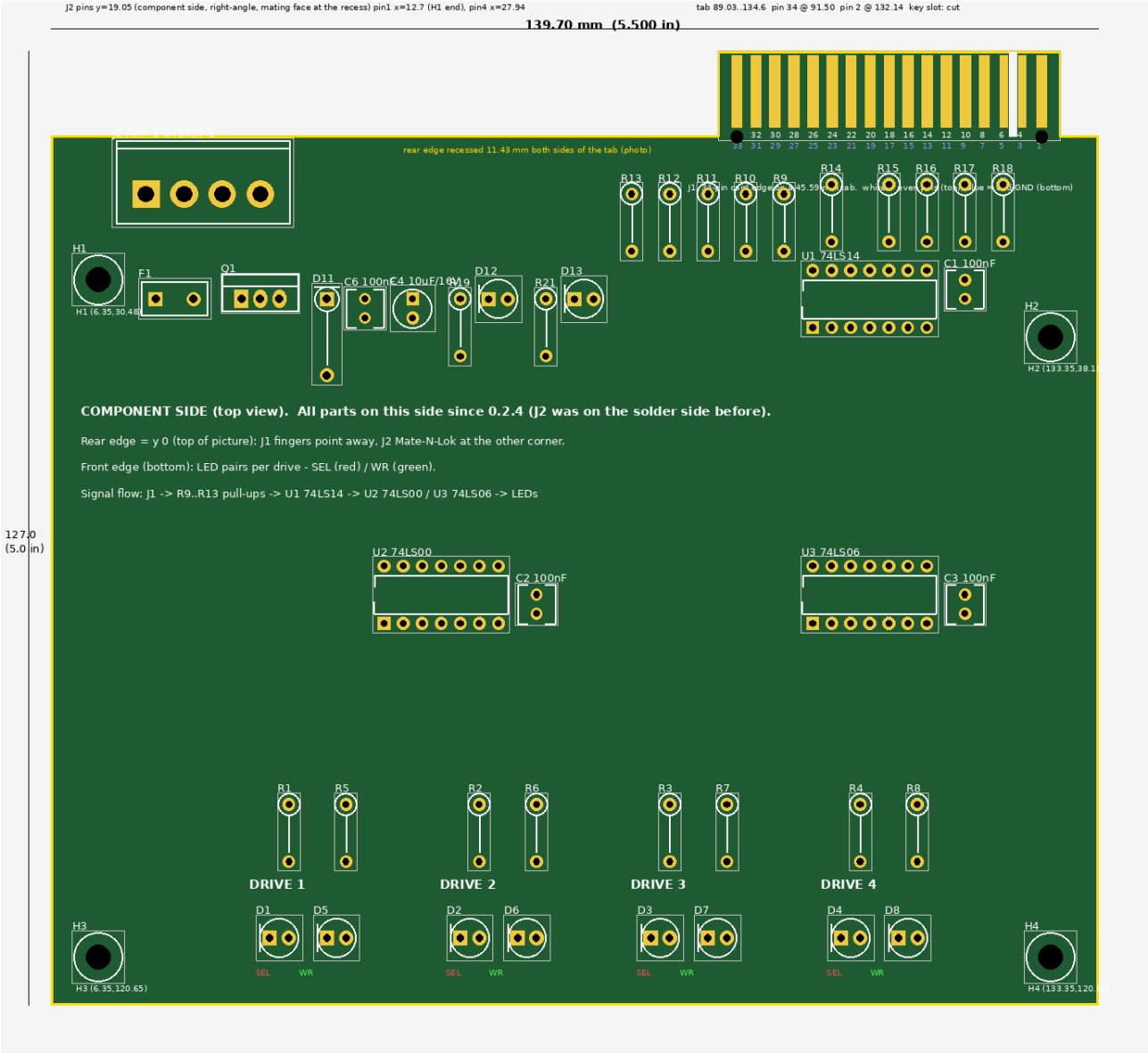
3.5 Reference: Ven’s SA400 PCB



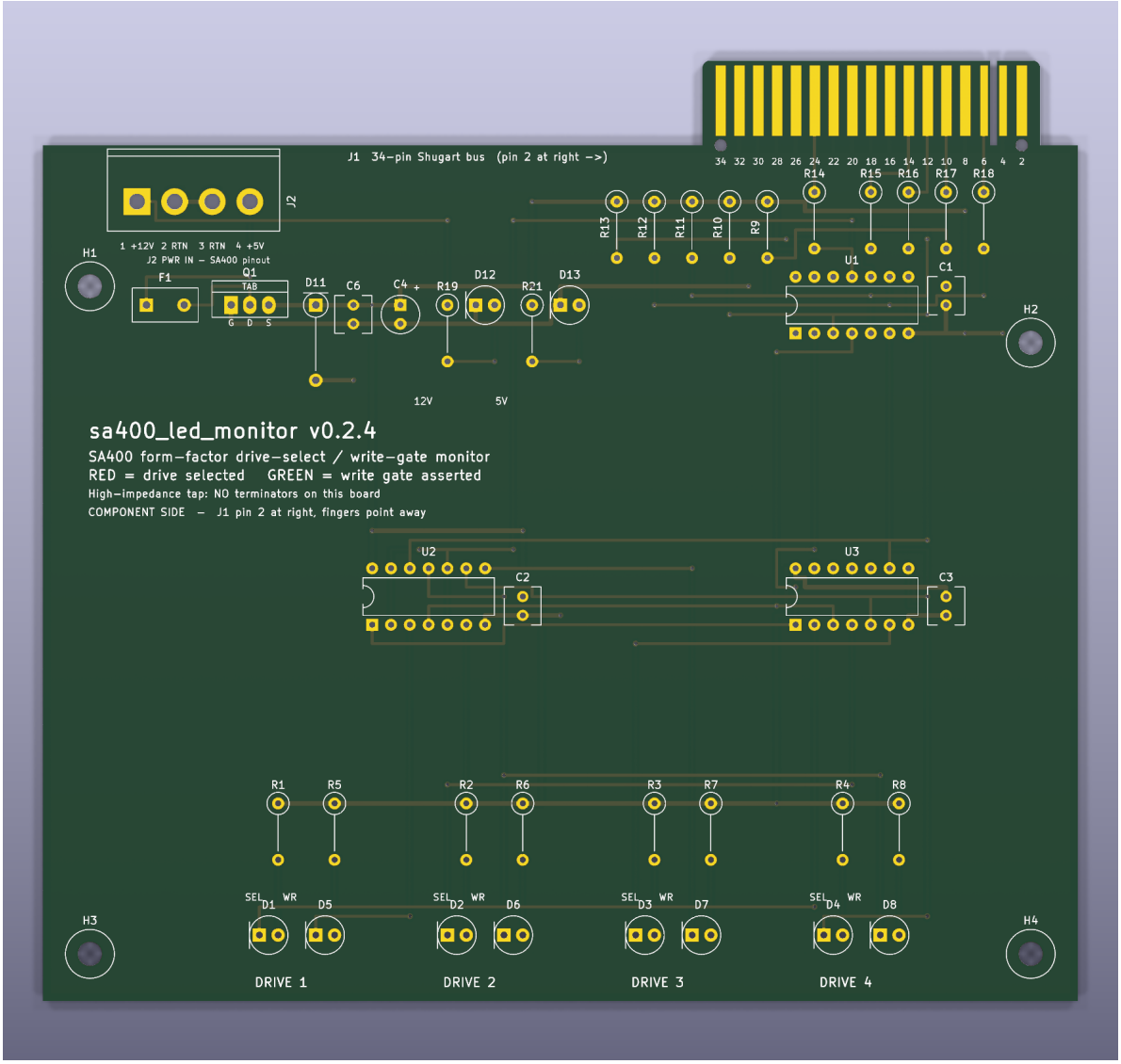


Left: the real tab. Right: this design, rotated to the same orientation.

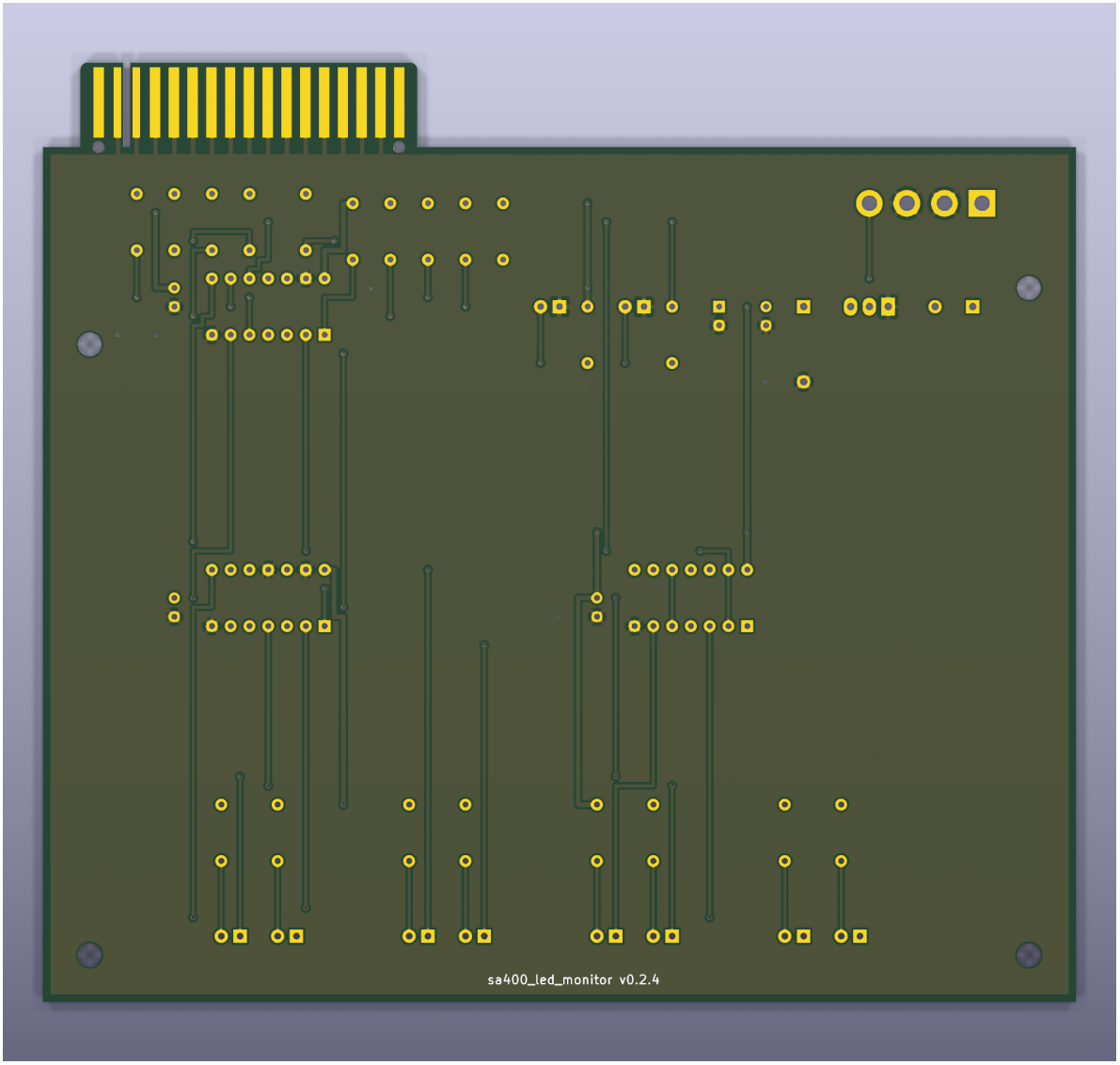
4 Placement and layout



Placement, component side (all parts on this side since 0.2.4).



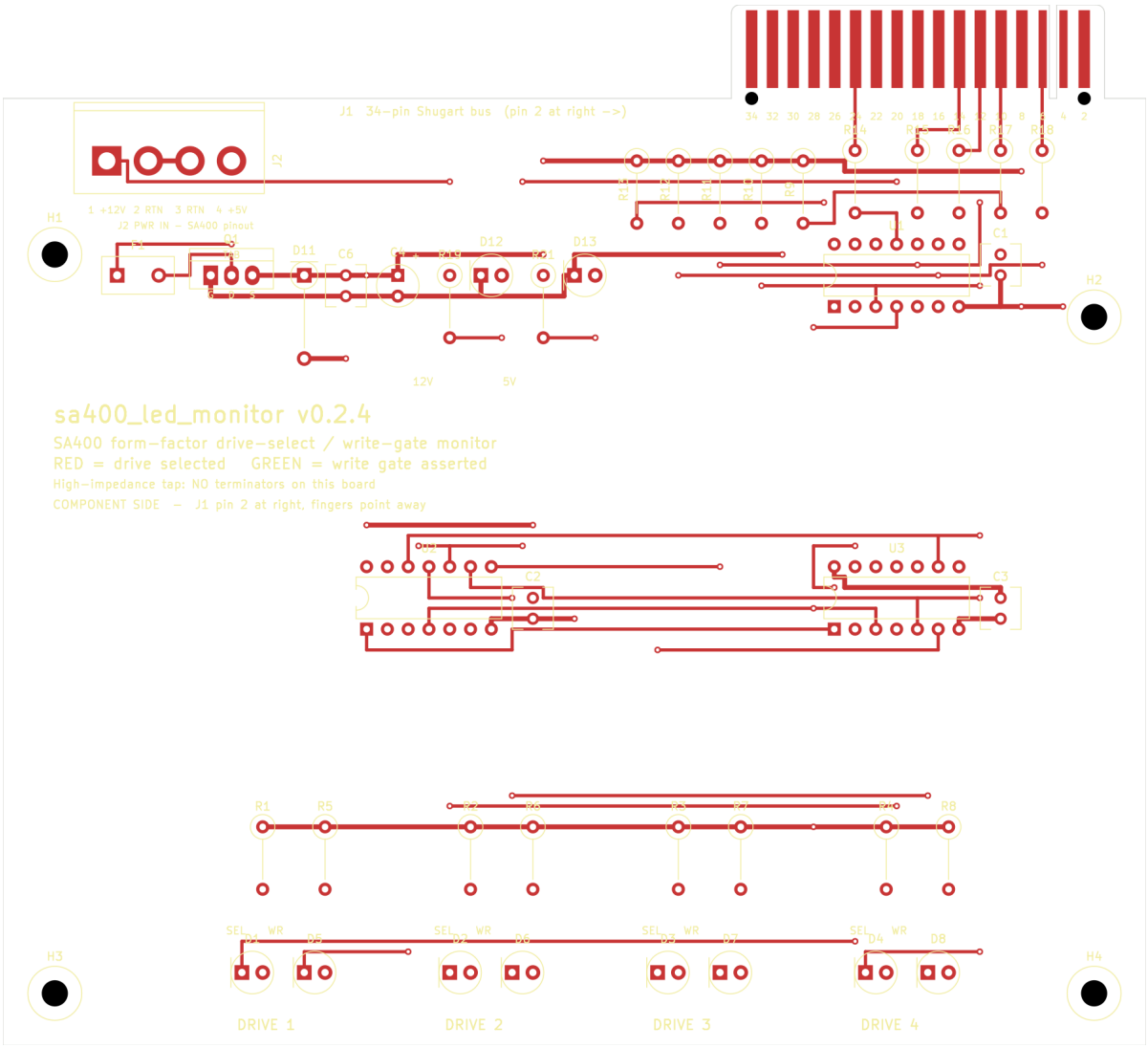
Component side (front), KiCad 10 raytraced render. J1 tab at the top right, J2 at the top left, LEDs along the front edge.



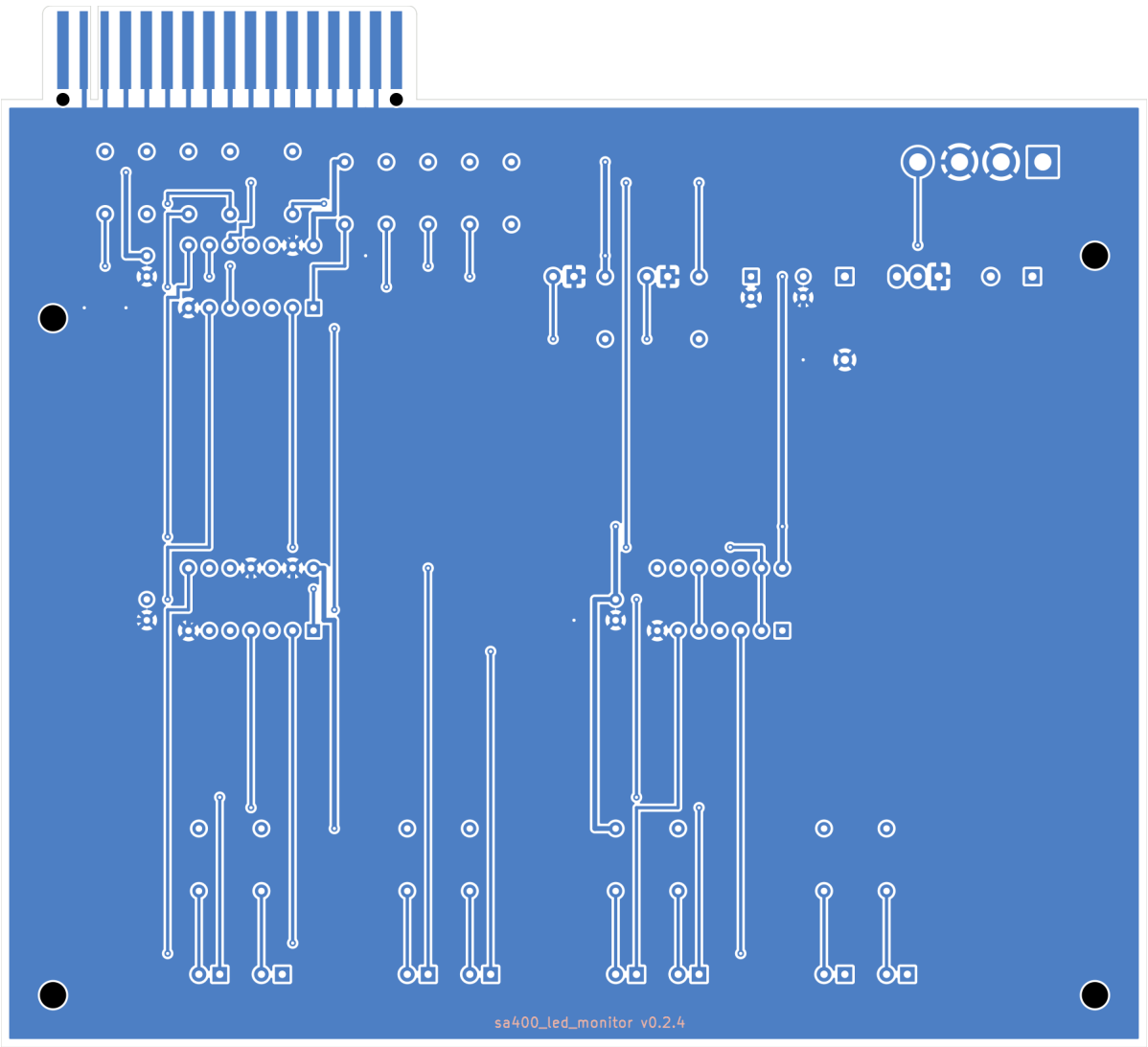
Solder side (back), seen from the back — so the J1 tab appears at the top *left* and J2 (the Mate-N-Lok header) at the top right. Olive areas are the filled GND pour; bare pads where the header, LEDs and DIPs solder.

4.1 Copper, side by side

The raytraced views above show the board under solder mask, which hides the copper. These are the copper layers themselves — what to look at when checking connectivity.

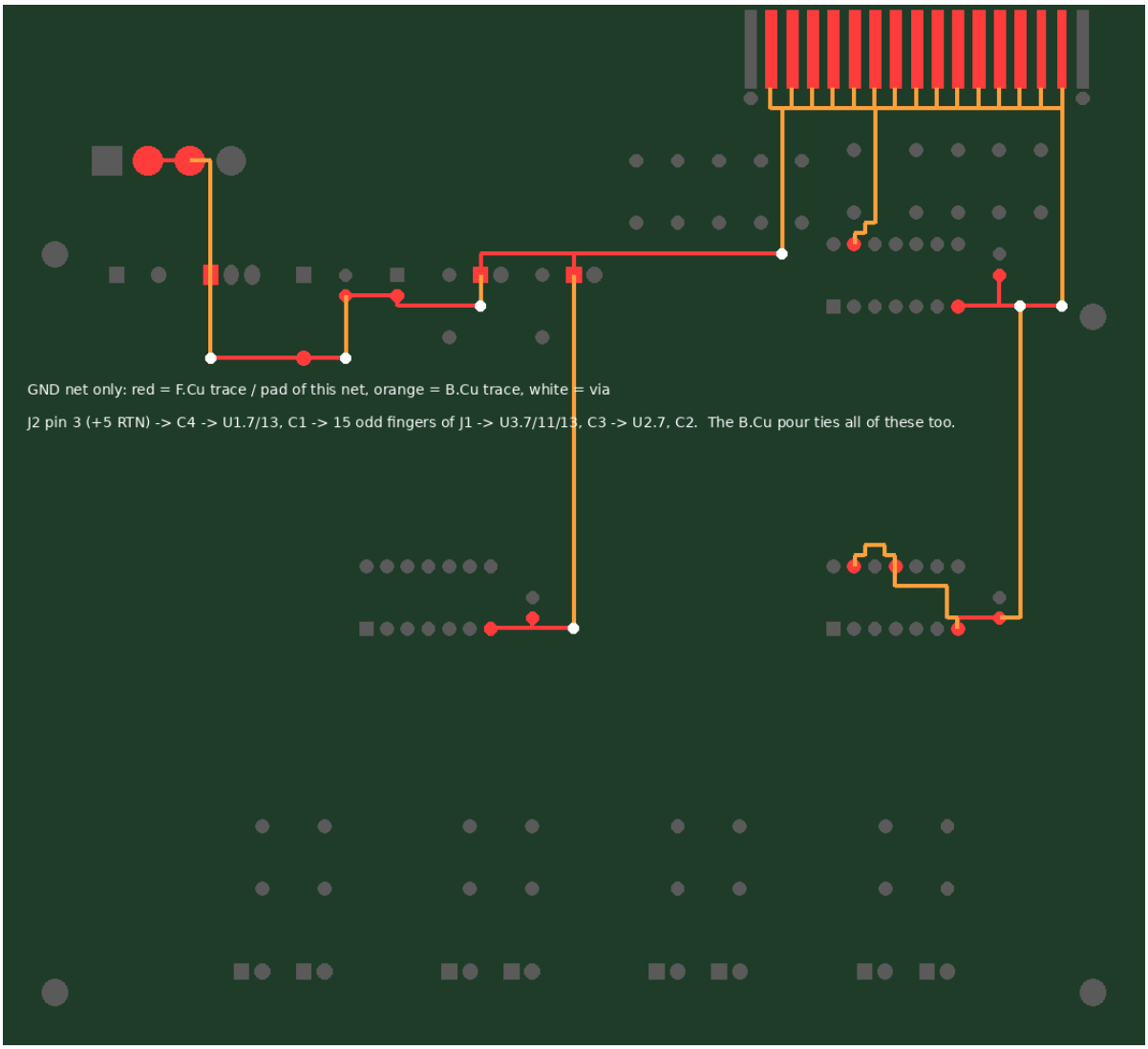


F.Cu — component-side copper: traces and pads (silkscreen overlaid for orientation).

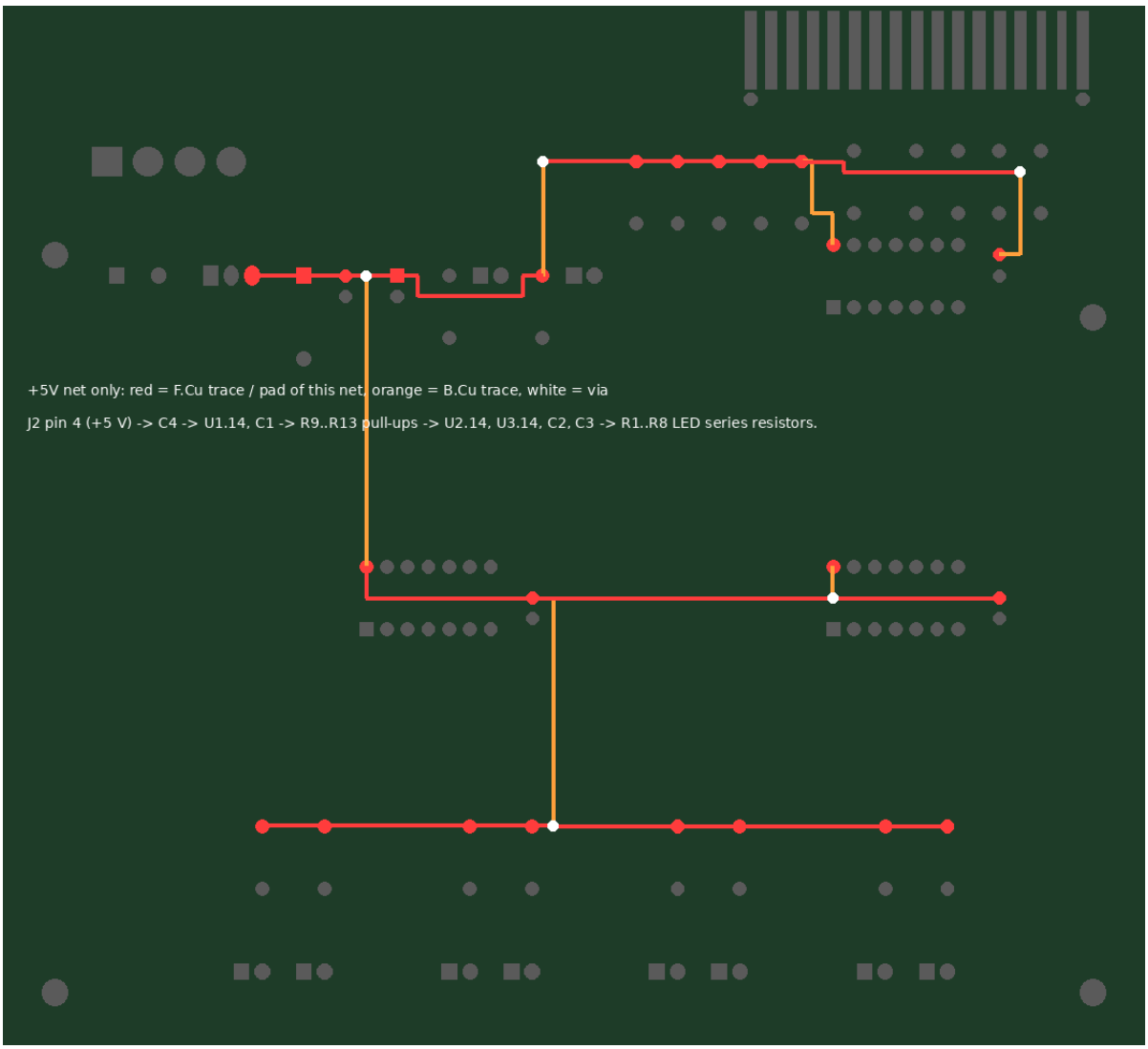


B.Cu — solder-side copper, viewed from the back (J2 at the top right). The blue field is the GND pour; every GND pad sits in it with thermal spokes, every other pad has a clearance ring.

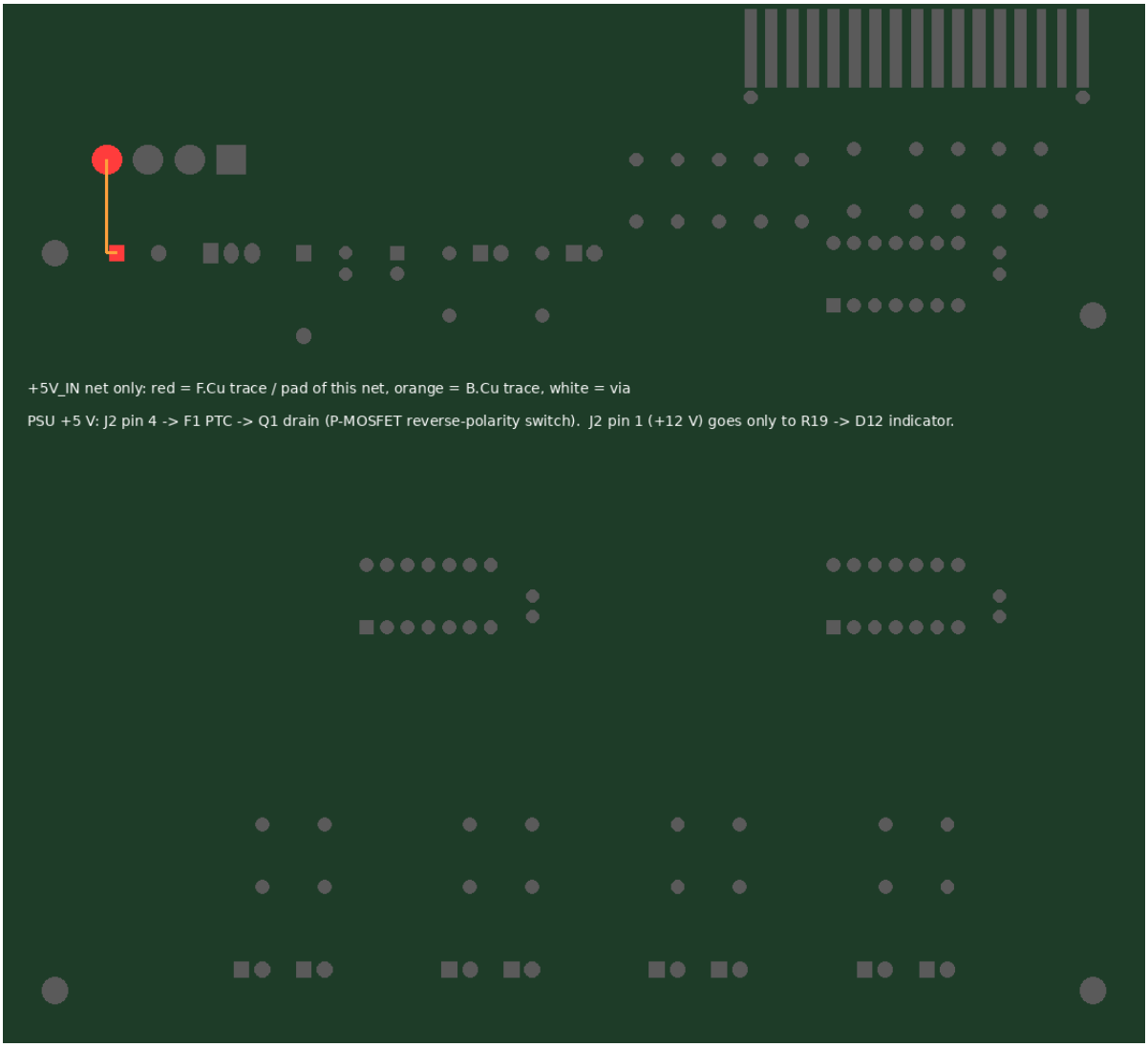
4.2 Power nets isolated



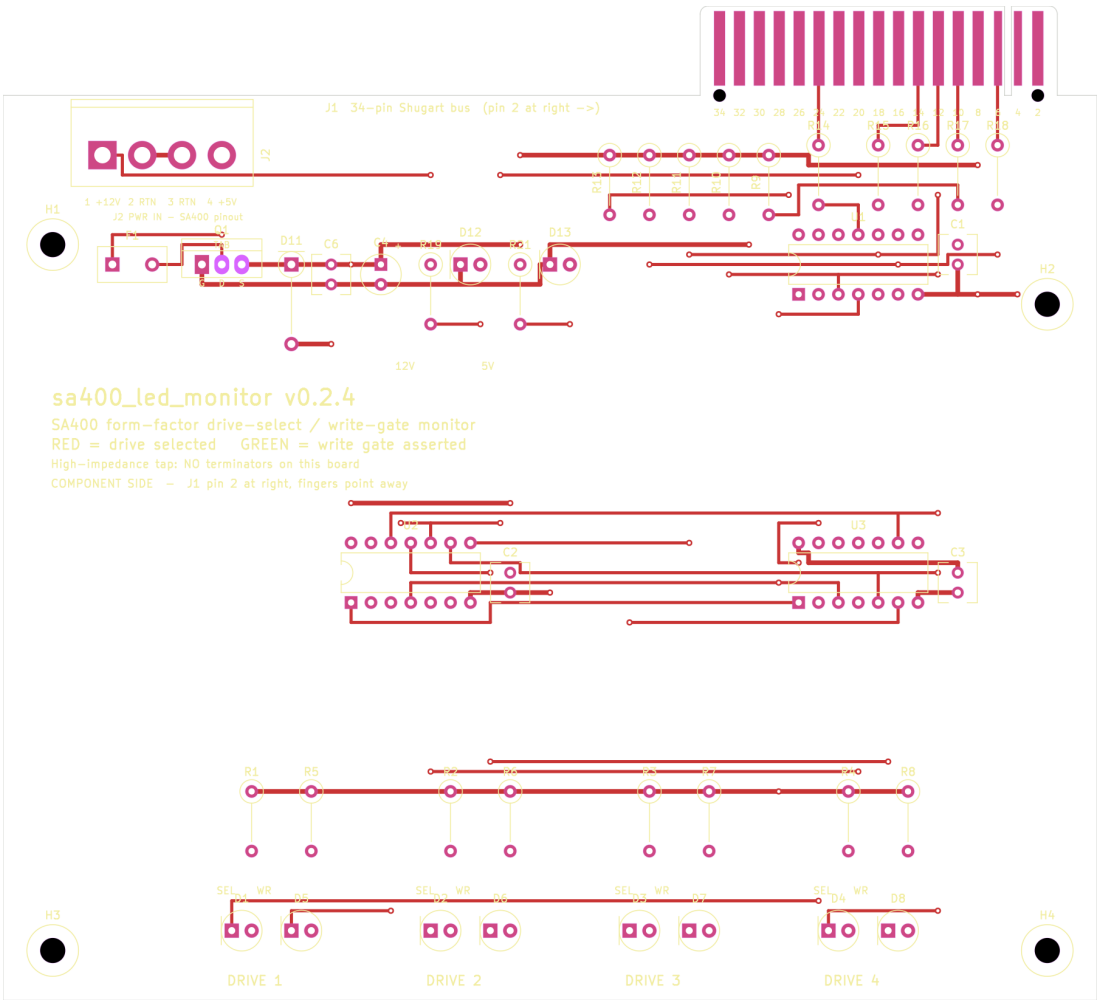
GND: J2 pins 2/3 → Q1 gate, D11, C6, C4, D12/D13 → U1 pins 7/13, C1 → the fifteen odd fingers of J1 → U3 pins 7/11/13, C3 → U2 pin 7, C2. The B.Cu pour ties all of these as well.



+5V: Q1 source → D11, C6, C4, R21 → U1 pin 14, C1 → pull-ups R9–R13 → U2/U3 pin 14, C2, C3 → LED series resistors R1–R8.



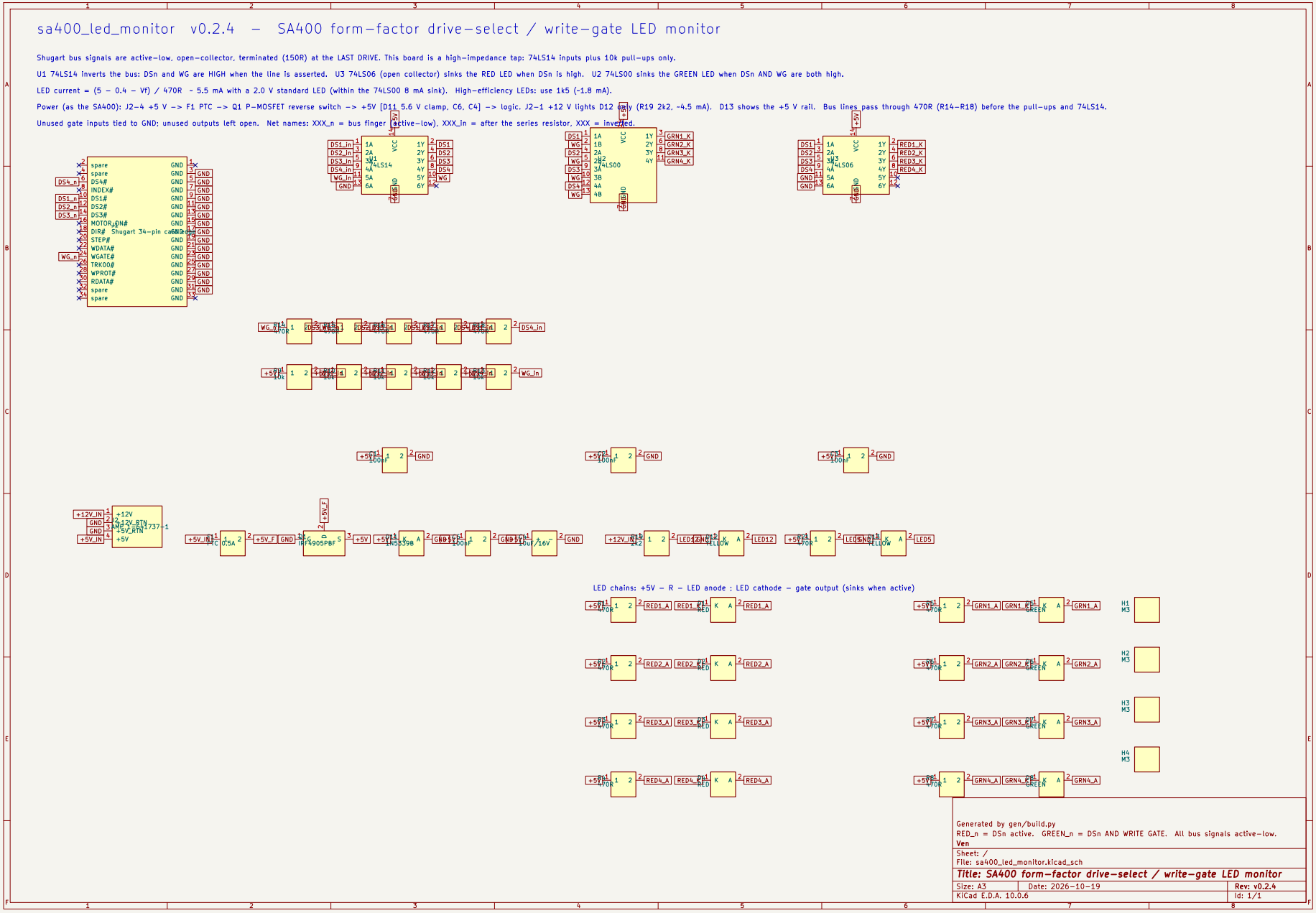
+5V_IN (PSU +5 V): J2 pin 4 → F1 → Q1 drain. J2 pin 1 (+12 V) goes only to R19 → D12.



Combined layer view (front copper red, back copper blue with the pour filled).

Design rules: 0.40 mm signal / 0.60 mm power traces, 0.30 mm clearance, 0.80/0.40 mm vias, 0.50 mm copper to edge (0.32 at the key slot, 0.66 at the fingers). JLCPCB minima are 0.127 / 0.127 / 0.45 / 0.30.

5 Schematic


sa400_led_monitor v0.2.4 — SA400 form-factor drive-select / write-gate LED monitor
Shugart bus signals are active-low, open-collector, terminated (150R) at the LAST DRIVE. This board is a high-impedance tap: 74LS14 inputs plus 10k pull-ups only.
U1 74LS14 inverts the bus: DSn and WG are HIGH when the line is asserted. U3 74LS06 (open collector) sinks the RED LED when DSn is high. U2 74LS00 sinks the GREEN LED when DSn AND WG are both high.
LED current = (5 − 0.4 − Vf) / 470R ≈ 5.5 mA with a 2.0 V standard LED (within the 74LS00 8 mA sink). High-efficiency LEDs: use 1k5 (≈1.8 mA).
Power (as the SA400): J2-4 +5 V -> F1 PTC -> Q1 P-MOSFET reverse switch -> +5V [D11 5.6 V clamp, C6, C4] -> logic. J2-1 +12 V lights D12 only (R19 2k2, ≈4.5 mA). D13 shows the +5 V rail. Bus lines pass through 470R (R14–R18) before the pull-ups and 74LS14.
Unused gate inputs tied to GND; unused outputs left open. Net names: XXX_n = bus finger (active-low), XXX_in = after the series resistor, XXX = inverted.
J2
Shugart 34-pin card-edge
spare
DS4#
INDEX#
DS1#
DS2#
DS3#
MOTOR_ON#
DIR#
STEP#
WDATA#
WGATE#
TRK00#
WPROT#
RDATA#
GND
U1
74LS14
U2
74LS00
U3
74LS06
LED chains: +5V — R — LED anode ; LED cathode — gate output (sinks when active)
H1
M3
H2
M3
H3
M3
H4
M3
Generated by gen/build.py
RED_n = DSn active. GREEN_n = DSn AND WRITE GATE. All bus signals active-low.
Ven
Sheet: /
File: sa400_led_monitor.kicad_sch
Title: SA400 form-factor drive-select / write-gate LED monitor
Size: A3
Date: 2026-10-19
Rev: v0.2.4
KiCad E.D.A. 10.0.6
Id: 1/1

Label-driven schematic as generated; every pin carries a global label, so the netlist is exact (KiCad 10 schematic parity: 0 issues). The full-size sheet is `output/sa400_led_monitor_schematic.pdf`.

6 Bill of materials

Qty	Ref	Part	Package	Example part / source
1	U1	74LS14 hex Schmitt-trigger inverter	DIP-14	TI SN74LS14N, LCSC C73064 https://www.ti.com/product/SN74LS14
1	U2	74LS00 quad 2-input NAND	DIP-14	TI SN74LS00N, LCSC C181629 https://www.ti.com/product/SN74LS00
1	U3	74LS06 hex inverter, open-collector	DIP-14	TI SN74LS06N, LCSC C22384767 https://www.ti.com/product/SN74LS06
3	U1A–U3A	DIP-14 socket	DIP-14	any dual-wipe 14-pin socket, LCSC C190815
1	F1	PTC 0.5 A hold / 1 A trip, $\geq$ 16 V	radial 5.08	A72-050 (LCSC C135425) or Littelfuse RUEF050 https://www.mouser.com/c/?q=RUEF050
1	Q1	P-MOSFET, pins G-D-S	TO-220 vertical, tab toward J2	IRF4905PBF (LCSC C2564) https://www.mouser.com/c/?q=IRF4905PBF; alt. onsemi NDP6020P
1	D11	Zener 5.6 V 5 W, +5 V clamp	DO-15 vertical	1N5339B (LCSC C5444832) https://www.mouser.com/c/?q=1N5339B
2	D12, D13	Yellow 5 mm LED, “12V” / “5V” present	5 mm	Kingbright WP483YDT (LCSC C6001325) https://www.mouser.com/c/?q=WP483YDT
4	D1–D4	Red 5 mm LED, standard brightness, $V_f \approx 1.9$ V	5 mm	Kingbright L-53ID (LCSC C598429) https://www.mouser.com/c/?q=L-53ID; high-efficiency WP7113LID also fits (then 1.5 kΩ)
4	D5–D8	Green 5 mm LED, standard brightness, $V_f \approx 2.1$ V	5 mm	Kingbright WP7113GD (LCSC C7117617) https://www.mouser.com/c/?q=WP7113GD; high-efficiency WP7113LGD also fits (then 1.5 kΩ)
8	R1–R8	470 Ω 5 % 1/4 W (1.5 kΩ for high-efficiency LEDs)	axial vertical 7.62	LED series, ≈ 5.5 mA; YAGEO MFR-25JT-52-470R, LCSC C176518
5	R9–R13	10 kΩ 5 % 1/4 W	axial vertical 7.62	bus pull-ups; YAGEO MFR-25JT-52-10K, LCSC C176451
5	R14–R18	470 Ω 5 % 1/4 W	axial vertical 7.62	bus series / fault-limiting; LCSC C176518
1	R19	2.2 kΩ 1/4 W	axial vertical 7.62	+12 V indicator series, ≈ 4.5 mA; YAGEO MFR-25FTE52-2K2, LCSC C172997
1	R21	470 Ω 1/4 W	axial vertical 7.62	+5 V indicator series, ≈ 6 mA; LCSC C176518
3	C1–C3	100 nF 50 V ceramic	disc 2.5 mm	IC decoupling; LCSC C45664

Qty	Ref	Part	Package	Example part / source
1	C4	10 μF 16 V electrolytic	radial 5 mm, 2.5 mm pitch	bulk on +5 V; + in the square pad; LCSC C53330112
1	C6	100 nF 50 V ceramic	disc 2.5 mm	rail entry decoupling; LCSC C45664
1	J2	TE Commercial Mate-N-Lok 4-pos right-angle header 1-641737-1	component side	https://www.te.com/en/product-1-641737-1.html; vertical 350211-1 (LCSC C305826) fits the same pads
—	J1	34-contact card edge, part of the PCB	—	gold fingers + bevel, fab option
4	H1–H4	M3 $\times$ 6 mm standoff + screw	—	any

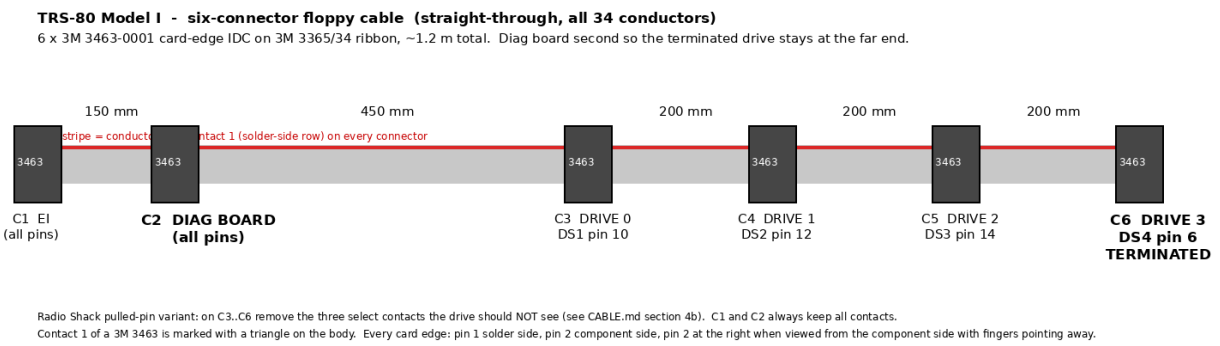
LED cathode = shorter lead / flat side of the rim, into the **square** pad (pin 1). Zener band (cathode) also goes in the square pad; Q1 gate is pin 1 (square) and its metal tab faces the J2 edge — the silkscreen shows the body, the tab line and G/D/S.

Mating parts (reference). 3M 3463-0001 34-way card-edge IDC (https://www.mouser.com/c/?q=3463-0001); 3M 3439-2 key plug for the pin 4/6 slot (https://www.mouser.com/c/?q=3439-2); AMP 1-480424-0 housing + 60619-1 pins for the power plug (https://www.te.com/en/product-1-480424-0.html).

Cost estimate (single board, 2026). ICs + sockets ≈ $4, protection and indicators $3, LEDs $2, passives $1, J2 $1, standoffs $1; PCB (JLCPCB, 5 pcs with the gold-finger option) $25–35 for the batch.

7 Six-connector Model I cable

Radio Shack's four-drive cable for the Model I has five 34-contact card-edge connectors: one for the Expansion Interface (EI) and one per drive. This cable adds a sixth for the monitor. Electrically the monitor is a high-impedance tap, so it can sit anywhere on the ribbon.



Why the monitor goes next to the EI, not at the end. The *last* drive on the cable carries the 150 Ω terminators (Radio Shack 26-1160 / 26-1164 are the terminated drives; 26-1161 is unterminated). Everything past the terminated drive is an unterminated stub, and a stub of any length is where reflections live. Putting the monitor second keeps the terminated drive at the physical end.

Parts. 6 × 3M 3463-0001 (https://www.mouser.com/c/?q=3463-0001); 1.3 m 3M 3365/34 ribbon (https://www.mouser.com/c/?q=3365%2F34); optional 3M 3448-56 strain reliefs; optional 3M 3439-2 key plug. The EI's disk connector is a 34-contact card edge like a drive's, and the monitor's tab is built to the same 3M drawing, so all six connectors are identical.

Drive select — two ways, pick one.

(a) Straight-through (recommended). All 34 contacts in every connector. Each drive is set to its own select with its own switch/shunt: drive 0 = DS1 = pin 10, drive 1 = DS2 = pin 12, drive 2 = DS3 = pin 14, drive 3 = DS4 = pin 6.

(b) Radio Shack "pulled pin" style. Radio Shack shipped Model I drives with *all four* selects enabled and pulled three select contacts from each drive connector, so cable position set the drive number: C3 keeps 10 (removes 12, 14, 6); C4 keeps 12; C5 keeps 14; C6 keeps 6. C1 (EI) and C2 (monitor) keep all contacts. Use this only if the drives are still shunted for all four selects — an all-select drive on a straight cable answers to every drive number and two drives fight on the data lines. To remove a 3M contact: before pressing the cover, pull the unwanted contact out of the body with fine pliers; the ribbon conductor above it stays intact for the drives further along.

Pin 1. Ribbon conductor 1 (red stripe) to contact 1 (triangle mark) of every connector. On every card edge pin 1 is on the solder side, pin 2 above it on the component side, pin 2 at the right viewed from the component side with the fingers pointing away.

Assembly. Cut and square the ribbon; mark 150 / 450 / 200 / 200 / 200 mm; lay it in each body with the stripe on the triangle side; pull contacts for (b); press the cover (3M 3640 press with the 3443-63 locator, or a vise with flat wooden blocks — press evenly, never rock); fit strain reliefs; buzz all 34 conductors end to end.

Check before first use. Terminated drive at C6, all other terminators removed; one select per drive; stripe on the same edge at all six connectors; monitor powered from its own J2. Power up with only the EI and the monitor connected: RED 1 should flash as the ROM tries to boot from drive 0, and no green should ever light without a write.

8 Fabrication — JLCPCB

8.1 Files

`output/gerbers/`: `-F_Cu.gtl`, `-B_Cu.gbl`, `-F_Mask.gts`, `-B_Mask.gbs`, `-F_Silkscreen.gto`, `-B_Silkscreen.gbo`, `-Edge_Cuts.gm1`, `-PTH.drl`, `-NPTH.drl`, two drill maps and the `.gbrjob`. Zip the folder and upload it as one archive.

8.2 Order options

Option	Choose	Why
Layers	2	
Thickness	**1.6 mm**	3M connector spec 1.37–1.78 mm
Copper	1 oz	
Surface finish	**ENIG**	required under the fingers; HASL is not acceptable on a card edge
Gold fingers	**Yes**	hard gold on the tab
Bevel / chamfer	**Yes**	JLCPCB bevels at 30°, 0.6 mm deep; finger copper starts 0.66 mm in to suit it
Solder mask / silk	any / white	
Castellated holes	No	

8.3 Verification already done (KiCad 10.0.6 `kicad-cli`)

Check	Result
`pcb drc --schematic-parity --severity-all`	**0 violations, 0 unconnected, 0 parity issues** (`output/drc.json`)
`sch erc --severity-all`	**0 violations** (`output/erc.json`)
Gerber / drill plot	`output/gerbers/`, 12 files
GND zone	filled and saved in the board file
J2 pin 1 on B.Cu	square pad at x = 27.94 in the plotted gerber

8.4 Before you upload

1. Open `kicad/sa400_led_monitor.kicad_pro` in KiCad 10.
2. Re-run DRC and ERC yourself: `gen/check.ps1` does both plus the gerber and drill plots.
3. Confirm the three measurements in Section 9; if any changes, rebuild (`python3 gen/build.py`, then `postprocess.py`) and re-plot.
4. Zip `output/gerbers/` and upload.

8.5 Assembly

The board is entirely through-hole with J2 on the back. JLCPCB's PCBA now hand/wave-solders THT parts ($3.50 setup + $0.0173 per joint) from their library or consigned

stock; PCBWay and Screaming Circuits offer turnkey assembly buying exact parts from any distributor. `output/assembly/` holds a JLCPCB-format BOM and CPL (LCSC column left blank for you to pick parts in their matcher) and the KiCad position file. For one to five boards, hand assembly (about 110 joints) is the sensible route and verifies the J2 hole size and LED polarity before paying an assembler. Details: `docs/ASSEMBLY.md`.

9 To do

v0.2.x boards (5 populated ordered from JLCPCB, re-ordered 2026-10-05 with the fab files only)

- ☐ Measure tab-edge-to-corner on the SA400 PCB → `J1_TAB_CORNER_GAP` (5.08 assumed)
- ☐ Measure key plug thickness (3M 3439-2) → `J1_KEY_SLOT_W` (0.91)
- ☐ Confirm the header post diameter → `J2_DRILL` (2.1; TE lists the 350211-1 post at 1.6 mm) — https://www.te.com/en/product-1-641737-1.html
- ☐ **Bench-check J2 handedness** (header in holes, plug on, yellow wire → `+12V` pad) on the 0.2.3 boards; sets `J2_PIN1_X`/`J2_PIN_DIR` for 0.2.4 — see docs/ASSEMBLY.md
- ☐ 0.2.4: J2 moved to the component side (right-angle 1-641737-1 or vertical 350211-1); single-sided PCBA now covers every part
- ☒ KiCad 10 DRC + ERC clean; schematic/PCB linked; F8 metadata emulated
- ☐ Run *Update PCB from Schematic* (F8) once in the GUI — expect an empty change list
- ☒ LED resistor decided: 470R (standard-brightness LEDs from JLCPCB stock); R19 2k2 — schematic/BOM/docs at 0.2.3
- ☒ JLCPCB BOM carries LCSC numbers; sockets listed as U1A–U3A with CPL rows (0.2.3)
- ☒ Q1 silkscreen redrawn to the TO-220 convention (tab toward J2) after JLCPCB's 2026-10-07 query (0.2.3)
- ☒ Reply to JLCPCB (orientation confirmed, 2026-10-07); boards received 2026-10-19
- ☒ Power protection: PTC, P-MOSFET reverse switch, 5.6 V clamp, series bus resistors (0.2.2)

Board bring-up — see docs/TEST_PLAN.md

- ☐ **J2 handedness check before any power** (TEST_PLAN to add): header in holes, drive plug on, continuity yellow wire → `+12V` pad; fit the header on whichever side passes
- ☐ Build the bus-simulator jig (TEST_PLAN §0.6)
- ☐ Board 1: Phase 1 → 2 (incl. protection tests 2.7/2.8) → 3
- ☐ Boards 2–5: Phase 1 → 2 (2.1–2.6) → 3 (3.3.1–3.3.4)
- ☐ TEST_PLAN 3.3.7 (design review F1): U1 input node on the terminated bus — ≤ 0.5 V clean; 0.5–0.8 V → fit a 74HCT14 in U1 (socket swap) and re-measure

☒ Update docs/BOM.md and THEORY_OF_OPERATION.md to the r2 as-built values (done in 0.2.3; design-review corrections merged 2026-10-19)

v0.3 — decided, see docs/V0.3_PLAN.md (starts after the v0.2.x test run)

☐ Four 8-bit common-cathode LED bar modules (one per drive) replace D1–D8

☐ 4 × 74HCT244 gated by the drive-select nodes; second 74LS14; 74LS123 for STEP/INDEX

☐ Six new bus lines: MOTOR ON, STEP, DIRECTION, TRACK 00, WRITE PROTECT, INDEX (each through 470 Ω)

☐ Confirm module colour split and resistor value (Ven)

☐ Design-review carry-overs (V0.3_PLAN): 74HCT14 receivers; pull-ups to the finger side of R14–R18, 100 kΩ; 1N4148 in series with D12; D11 value per 2.4.2

☐ Optional plated variant of the clip holes (deferred)

Tooling

☐ Router: courtyard-aware silkscreen placement

☐ Router: DSN export so freerouting can be used for denser boards

☐ Rebuild `pdf/` (LuaLaTeX) whenever docs change — `python pdf/build_pdf.py`

10 Changelog

v0.2.4 — 2026-10-19 — J2 on the component side

- **J2 moved to the component side**, right-angle, mating face at the recessed rear edge (TE Commercial Mate-N-Lok 1-641737-1; the vertical 350211-1 / LCSC C305826 fits the same pads and courtyard). Reason: JLCPCB's Economic PCBA is single-sided placement, and J2 was the only bottom-side part on the 0.2.3 order, so it came back unpopulated. All 45 placements are now Top.
- Pin 1 at the H1 (corner) end, x = 12.70, pins toward the board centre — the mirror of the solder-side layout, same ASSUMED header handedness; `J2_PIN_DIR` added so the end can be flipped from config. The bench continuity check (yellow wire → `+12V` pad) is now a TODO item and an ASSEMBLY.md step.
- Power row (F1, Q1, D11, C6, C4, R19, D12, R21, D13) moved from y 30.48 to 33.02 so the J2 labels (`1 +12V … 4 +5V`, "J2 PWR IN – SA400 pinout", below the housing) clear the Q1/F1 references; "12V"/"5V" silk moved with it. Component-side J2 pin-number/function labels of 0.2.2–0.2.3 and the solder-side labels are gone (nothing on the back now but the GND pour and the version text).
- Footprint renamed `MateNLok_4_RA` (was `MateNLok_4_RA_350211-1`); J2 value `AMP 1-641737-1`; JLCPCB BOM leaves J2's LCSC blank (consign the right-angle part or match C305826 for the vertical one).

- Docs: THEORY §4.0 (why moving a header between sides mirrors pin 1), MECHANICAL, ASSEMBLY (handedness check), FABRICATION (Economic tier is enough), BOM, TODO. Copper re-routed by the generator (power row moved); DRC/ERC/links/F8 emulation re-verified.
- Merged from the bring-up session (same day, worked on the 0.2.2 tree): - Docs (2026-10-19): `docs/TEST_PLAN.md` added — three-phase bring-up plan (unpowered, bench supply with bus-simulator jig and protection tests, on the Model I EI), written against the r2 as-built BOM. - Docs (2026-10-19, test plan 1.1): §0.7 populated-board wireframe + legend; `gen/render_populated.py` draws it from `design.py`; `pdf/build_test_plan.py` builds the test-plan PDF (LuaLaTeX via pandoc). - Docs (2026-10-19, design review, test plan 1.2): receiver low-level margin found thinner than documented — THEORY §3.0 omitted the pull-up current through the 470 Ω (worst case ≈ 0.79 V at U1 vs V_T− 0.5 V min); §3.0/§4.2/§4.3 corrected in red. TEST_PLAN §0.8 lists the findings, adds 3.2.5/3.3.7 (U1 input node on the real bus), loosens 2.6.17 to < 0.45 V, and marks 2.7.2/2.7.4 as able to damage D12/D11. V0.3_PLAN gains the carry-over fixes: 74HCT14 receivers, pull-ups on the finger side (100 k), 1N4148 with D12. `pdf/color_spans.lua` renders the red/blue Markdown spans in the test-plan PDF. - Docs (2026-10-19, test plan 1.3): drawings moved out of the front matter into appendices — A board wireframe (+ legend on its back), B one-page schematic (`pdf/schematic.tex`, circuitikz, as-built values and DIP pin numbers), C as-built parts list; each appendix starts on a right-hand page for double-sided printing. Design-review section renumbered §0.8 → §0.7.

v0.2.3 — 2026-10-07 — Q1 silkscreen, as-built values, JLCPCB BOM

- **Q1 footprint:** the TO-220 silkscreen box and courtyard were drawn on the +y side of the pin row; a vertical TO-220 with pin 1 at −x stands over the pins with its metal tab at −y (Vishay TO-220 drawing, KiCad `TO-220-3_Vertical`). JLCPCB's engineer queried the mismatch on their 3D render (2026-10-07); their render was electrically correct. Silk now shows the body outline, the tab line, "TAB", and G / D / S under the pads; F.Fab body added; courtyard y −3.4…+1.9. The component-side "J2 PWR IN" label moved from y 25.4 to 24.3 to clear the relocated Q1 reference. Copper, drills and outline unchanged.
- **As-built values:** R1–R8 and R21 1k5 → **470R**, R19 4k7 → **2k2** (standard-brightness LEDs from JLCPCB stock: L-53ID, WP7113GD, WP483YDT); Q1 value NDP6020P → **IRF4905PBF** (the G-D-S part JLCPCB stocks; NDP6020P stays the logic-level alternative). Schematic, BOM and theory of operation updated; +5 V budget ≈ 45 mA, +12 V indicator 4.5 mA.
- **JLCPCB BOM/CPL:** `build.py` now writes the LCSC numbers JLCPCB matched for the first order (`JLC` table) and lists the sockets as U1A/U2A/U3A; `verify.py` adds the three socket rows to the CPL at the IC positions, checks BOM ↔ CPL (45 designators) and zips the gerbers as `output/sa400_led_monitor_v0.2.3_gerbers_JLCPCB.zip`.
- Docs: FABRICATION (upload only the three files; order options; remark; how to answer an engineering query), ASSEMBLY (polarity table), BOM (LCSC column), MECHANICAL (Q1 geometry), TODO.
- History: the first PCBA order (2026-09-28) was cancelled because the whole project zip had been uploaded as the gerber file; re-ordered 2026-10-05 with the gerber zip + hand-edited r2 BOM/CPL, which this release reproduces from the generator.

v0.2.2 — 2026-09-08 — J2 labelling

- Docs (later the same day): `docs/V0.3_PLAN.md` records the decided next board (LED-bar modules, 74HCT244 per drive, TTL); TODO.md v0.3 section updated; JLCPCB upload zip added to `output/`.
- Fix: `output/assembly/BOM_jlcpcb.csv` and `CPL_jlcpcb.csv` had been left over from v0.1 (34 parts, old positions). BOM is now written by `build.py`, CPL by `verify.py` from KiCad's position export, and the gate checks the two list the same designators (42 placements + 3 sockets).
- J2 now labelled on both sides: component side has pin numbers 1–4 above the pads and +12V / RTN / RTN / +5V below them; solder side (footprint silk, outside the housing) has "1 +12V 2 RTN 3 RTN 4 +5V" and "PWR IN - SA400 pinout - pin 1 square".
- Fix: the solder-side housing outline was mirrored (drawn behind the pins instead of toward the rear edge). Back-side geometry is now pre-mirrored so KiCad's flip lands it where designed; the library copy follows the same convention (library parity still clean).

v0.2.1 — 2026-09-08 — SA400 power usage restored, protection kept

- Logic powered from the PSU **+5 V** pin as on the SA400; +12 V pin lights a "12 V present" indicator. Chain: J2-4 → F1 PTC 0.5 A → Q1 NDP6020P P-MOSFET reverse-polarity switch (0.002 V drop) → +5 V with D11 1N5339B 5.6 V clamp, C6, C4. D13 "5 V" and D12 "12 V" indicators.
- Removed the 7805, 1N5819, P6KE18A, C5, C7 of v0.2.0. Series 470 Ω bus resistors retained.
- Fault table updated (THEORY §4.3). 47 footprints; DRC 0/0/0, ERC 0, links 47/47, F8 emulation clean.
- Six markdown docs (README, BOM, MECHANICAL, FABRICATION, ASSEMBLY, TODO) had been shipped empty since v0.1.3 by a versioning bug; all rebuilt, and `gen/verify.py` now fails on an empty doc.

v0.2.0 — 2026-09-08 — regulated power and bus protection

- Board now powered from **+12 V** (J2 pin 1) through F1 PTC 0.5 A → D9 1N5819 (reverse) → VIN (D10 P6KE18A TVS, C5 10 μF/25 V, C6 100 nF) → U4 7805 → +5 V (C7 100 nF, D11 1N5339B 5.6 V clamp, C4 10 μF). PSU +5 V pin unused; both returns to GND.
- 470 Ω series resistors R14–R18 between each monitored finger and its pull-up/receiver node (nets `DSn_n` = finger, `DSn_in` = receiver side). Worst-case current into an EI driver $\leq$ 11 mA.
- Fault analysis in THEORY_OF_OPERATION §4; BOM, schematic, placement, routing regenerated.
- 47 footprints; KiCad 10 DRC 0/0/0, ERC 0, links 47/47, F8 emulation clean.

v0.1.4 — 2026-09-08

- F8 (*Update PCB from Schematic*) on v0.1.3 matched every part but still listed three metadata updates per footprint: sheetname, sheetfile and ”unit metadata” (KiCad 10’s per-unit pin list used for gate swapping). `gen/postprocess.py` now writes all three, taking the unit/pin lists from KiCad’s own netlist export so they compare equal in the updater.
- `gen/verify.py` step 5b emulates the updater’s dry-run comparisons (footprint id, value, sheetname, sheetfile, unit pin lists) from the exported netlist; a negative test confirms it flags a board that lacks the metadata.
- Board file otherwise unchanged; all outputs regenerated.

v0.1.3 — 2026-09-07

- **Fix:** footprint links used `/<root-sheet-uuid>/<symbol-uuid>`; KiCad’s root-sheet form is `/<symbol-uuid>` (checked against KiCad 10’s own `demos/ecc83`). v0.1.2 therefore still offered to add every part in *Update PCB from Schematic*. Corrected; `gen/verify.py` now checks this form.
- `gen/verify.py` added: build → KiCad postprocess → DRC+parity → ERC → link check → gerber/drill plot, nonzero exit on any failure. Nothing is packaged unless it passes.
- Not headlessly testable: the F8 dialog itself — run it once after opening the project.
- Docs (2026-09-08): per-side copper views (`output/copper_top.png`, `copper_bottom.png`) and isolated GND / +5V net images (`output/net_gnd.png`, `net_5v.png`, from `gen/render_nets.py`) added to the design PDF; the raytraced renders hide copper under the mask and had prompted a false ”no ground” read.

v0.1.2 — 2026-09-07

- Footprints now carry `(path "/<root>/<symbol-uuid>")` links to deterministic schematic symbol UUIDs (`uuid5(ref)`), so **Update PCB from Schematic** reports no changes instead of trying to add every part again. Verified: 34/34 linked, KiCad 10 DRC parity 0, ERC 0.
- `docs/ASSEMBLY.md` + `output/assembly/` (JLCPCB BOM/CPL, KiCad position file).
- All fab outputs, renders and the design PDF regenerated as v0.1.2.

v0.1.1 — 2026-09-07 Verification pass with KiCad 10.0.6 (kicad-cli DRC/ERC, pcbnew API).

- Finger copper set back 0.66 mm from the insertion edge (3M .026; JLCPCB bevel rule $\geq$ 0.6 mm).
- Pull-ups spaced to 5.08 mm, LED pairs to 7.62 mm (courtyards were overlapping).
- No-connect pads carry KiCad’s `unconnected-(…)` nets → schematic parity exact.
- All footprints moved into the project library `sa400.pretty`; symbols into `sa400.kicad_sym` with project lib tables.

- J2 written front-side and flipped by KiCad itself (`gen/postprocess.py`), which also fills the GND zone and saves the board in native KiCad 10 format.
- J2 pins 3/4 typed power_out so ERC sees the supply; symbol placement snapped to 1.27 mm.
- Silkscreen: 0.8 mm text, cap outlines off the pads, labels de-overlapped.
- Result: DRC 0 / unconnected 0 / parity 0, ERC 0. Gerbers, drills, STEP re-exported from 10.0.6.

v0.1.0 — 2026-09-05 First complete design.

- Board 5.5 × 5.0 in, rear edge recessed 11.43 mm, J1 tab per 3M TS-0010-14 (45.59 mm, 1.40 mm contacts, 10.16 mm long, R1.02 corners), key slot 0.91 × 11.43 between pins 4/6, two Ø1.6 NPTH clip holes on the root line under pins 2 and 34, pin 2 5.08 mm from the corner.
- J2 AMP 350211-1 on the solder side, +5 V only.
- 74LS14 receivers + 10k pull-ups, 74LS06 → red (DSn), 74LS00 → green (DSn·WG), 1k5 LEDs.
- Generated KiCad 7-format project (opens in 9), validated by kicad-cli 7.0.11: gerbers and drills plotted, schematic netlist matches PCB 111/111 pins.
- Python maze router, 0.30 mm clearance, independent geometric DRC clean.
- Docs: THEORY_OF_OPERATION, BOM, CABLE (6-connector Model I cable), MECHANICAL, FABRICATION.

1. notches → full-length rear recess; 2. all 34 fingers restored; 3. key slot cut, full finger depth; 4. tab width 45.59 per 3M; 5. clip holes moved onto the root line, non-plated.

11 References

1. Shugart SA400L Minifloppy OEM Manual, P/N 39019-1, Nov 1982 — §3.2 pinout, §3.2.1 electrical, §3.3 J2, fig 3-2 card edge.
   https://www.bitsavers.org/pdf/shugart/SA400/39019-1_SA400L_OEM_Manual_Nov82.pdf
2. Shugart SA400 Service Manual, P/N 54096-2, Apr 1979 — schematics, parts catalogue (PCB clamps 54036).
   https://archive.org/details/bitsavers_shugartSA4eApr79_4138915
3. 3M Card-Edge .100 in Connector, TS-0010-14 — recommended PCB layout for the 3463.
   https://multimedia.3m.com/mws/media/220130/3m-tm-100-in-card-edge-connector-w-o-mounting-fl
   pdf
4. TEAC FD-55 Specification — fig 103 card-edge dimensions (cross-check).
   https://deramp.com/downloads/floppy_drives/teac/TEAC%20FD-55GFR.pdf
5. JLCPCB gold-finger design rules. https://jlcpcb.com/help/article/jlcpcb-gold-fingers

6. TI SN74LS14 / SN74LS00 / SN74LS06 datasheets. https://www.ti.com/product/SN74LS14
7. KiCad 10.0.6. https://www.kicad.org/download/

A Appendix: gen/config.py

Every dimension used by the generator, with its source.

```
"""
Design configuration for the SA400 form-factor LED monitor board.

Every dimension is in millimetres unless the name says otherwise.
Every value that came from a document has its source in the comment.
Anything marked ASSUMED is a design decision that must be verified
against hardware before ordering (see docs/MECHANICAL.md).
"""

VERSION = "0.2.4"
PROJECT = "sa400_led_monitor"
TITLE = "SA400␣form-factor␣drive-select␣/␣write-gate␣LED␣monitor"

IN = 25.4

# ------------------------------------------------------------- board
BOARD_W = 5.5 * IN # 139.70 user spec: 5.5 in edge carries J1 and J2
BOARD_H = 5.0 * IN # 127.00 user spec: 5.0 in (5.5 in is also acceptable)
BOARD_THICKNESS = 1.6 # SA400L OEM manual fig 3-2: 0.062 +/- 0.007 in

# ------------------------------------------------------------- J1 card edge
# SA400L OEM manual (P/N 39019-1) section 3.2 and figure 3-2
J1_PITCH = 0.100 * IN # 2.54 contact pitch
# 3M TS-0010-14 sheet 3 "Recommended PCB Layout" for the 3463 (34-way) connector is THE standard
# for this edge; Shugart fig 3-2 copies it. https://multimedia.3m.com/mws/media/220130/3m-tm-100-in-card-
    edge-connector-w-o-mounting-flng-ts0010.pdf
J1_FINGER_W = 0.055 * IN # 1.40 3M: contact width .055 +/- .005
J1_FINGER_LEN = 0.400 * IN # 10.16 3M: contact reach .400 +/- .015 from the insertion edge
J1_FINGER_SETBACK = 0.66 # 3M ".026 [0.66]" copper setback from the edge; JLCPCB needs >= 0.6 for the
    bevel
J1_FINGER_DEPTH = 0.450 * IN # 11.43 3M: tab depth .450 MIN (Shugart fig 3-2 0.450)
J1_TAB_W = 1.795 * IN # 45.59 3M dimension A (+/- .005) = Shugart fig 3-2 1.795
J1_TAB_MARGIN = (J1_TAB_W - 16 * J1_PITCH - J1_FINGER_W) / 2 # 1.775 bare board beyond pins 2 and 34
J1_PIN2_FROM_RIGHT_EDGE = J1_TAB_W / 2 - 8 * J1_PITCH # 2.475: pin 2 centre to tab edge
# TEAC FD-55 spec fig 103 shows the same construction: fingers on a 45.44 mm tab, key slot 0.91.
# PHOTO of a real SA400 PCB (P/N 25080, supplied by Ven, 2026-09-05): the tab PROTRUDES from the
# rear edge - the whole rear edge is set back by the tab depth on both sides, the tab sits about
# 5 mm from the pin-2 corner, two small holes sit at the tab root in line with pins 2 and 34,
# and that particular board has NO key slot cut (keying is "optional" per the OEM manual).
J1_REAR_RECESS = J1_FINGER_DEPTH # 11.43 rear edge set back by the tab depth (photo)
J1_TAB_CORNER_GAP = 5.08 # ASSUMED +/-1.5: pin-2 side shoulder to the board corner, scaled from the photo
J1_TAB_RIGHT = BOARD_W - J1_TAB_CORNER_GAP
J1_KEY_SLOT = True # key slot between pins 4 and 6 (photo + OEM manual sec 3.2)
J1_POST_HOLES = True # 2 x 1.6 mm holes at the tab root in line with pins 2 and 34 (photo, fig 3-2 "0.063
     NOM (2X)")
J1_POST_HOLE_DIA = 1.6
J1_TAB_CORNER_R = 1.02 # 3M: R.040
J1_POST_HOLE_Y = J1_FINGER_DEPTH # 11.43: hole centred on the tab-root line, ring touching the finger end
     (photo, scaled off the 2.54 pitch)
J1_KEY_SLOT_DEPTH = J1_FINGER_DEPTH # 11.43: full finger length, per Ven's board (fig 3-2 reads 0.36 in)
J1_KEY_SLOT_W = 0.91 # 3M: .036 +/- .004 (TEAC fig 103 agrees)
J1_KEY_BETWEEN = (4, 6) # section 3.2: key slot between pins 4 and 6
J1_NARROW_PADS = {3: 1.0, 4: 1.0, 5: 1.0, 6: 1.0} if J1_KEY_SLOT else {} # 0.32 mm copper-to-slot when
    cut
J1_DROP_PADS = set() # all 34 contacts present
# Handedness (established in the extender / activity-monitor projects):
# viewed from the COMPONENT side with the card edge pointing away, pin 2 is at the RIGHT.
# Even pins on the component side (F.Cu), odd pins on the solder side (B.Cu).

# ------------------------------------------------------------- J2 power
# SA400L OEM manual section 3.3: AMP Mate-N-Lok 4 pins, pin 1 = +12 V, 2 = +12 RTN, 3 = +5 RTN,
```

```
# 4 = +5 V. The SA400 carries it on the solder side; v0.2.0-0.2.3 copied that and JLCPCB's
# single-sided (Economic) PCBA left it unpopulated. From v0.2.4 J2 is on the COMPONENT side,
# right-angle, mating face at the recessed rear edge (Ven, 2026-10-19).
# Part: TE Commercial MATE-N-LOK right-angle 4-pos header 1-641737-1 (ENG_CD_641737); the
# vertical 350211-1 (LCSC C305826, termination post 1.6 mm [.063], mating pin 2.1 mm) fits the
# same pads and courtyard if the right-angle part is not in stock.
J2_PITCH = 5.08 # 0.200 in TE centreline
J2_DRILL = 2.1 # TE post 1.6 mm -> 0.25 mm clearance; ASSUMED until the drawing is checked
J2_PAD = 3.6
J2_ROW_Y = 19.05 # pin row 7.62 mm behind the recessed rear edge - ASSUMED
J2_LAYER = "F.Cu"
# Pin-1 end. Moving a header from the solder side to the component side MIRRORS it: the
# bottom-side layout (0.2.0-0.2.3) had pin 1 at x = 27.94 (nearest the board centre) viewed from
# the top; the same header on the top side puts pin 1 at the H1 end. Both rest on the ASSUMED
# handedness of the TE header (pin 1 on the header's left when standing on its tails, mating face
# away from you). BENCH CHECK before power (docs/TODO.md): header in the holes, drive plug on,
# continuity yellow wire -> the pad marked +12V. If it lands on +5V instead, set J2_PIN1_X to the
# other end (27.94 with J2_PIN_DIR = -1) and rebuild.
J2_PIN1_X = 12.70 # pin 1 at the H1 end for a component-side header - ASSUMED (see above)
J2_PIN_DIR = +1 # +1: pins 2..4 run toward the board centre; -1: toward the corner

# ------------------------------------------------------------------ mounting
MOUNT_HOLE_DRILL = 3.2 # M3 / #4-40 clearance. NOTE: the real SA400 PCB is held by
MOUNT_HOLE_PAD = 6.0 # four snap clamps (P/N 54036), not screws. Holes are a
MOUNT_HOLES = [ # deliberate deviation for a standalone board.
   (6.35, 30.48), (133.35, 38.10), (6.35, 120.65), (133.35, 120.65),
]

# ------------------------------------------------------------------ fab rules (JLCPCB 2-layer)
TRACE_W = 0.4
POWER_W = 0.6
CLEARANCE = 0.3
VIA_DRILL = 0.4
VIA_DIA = 0.8
EDGE_CLEARANCE = 0.5 # copper to board edge; JLC minimum 0.3
GRID = 1.27 # routing grid

# ------------------------------------------------------------------ LEDs
LED_R = 470 # ohms: ~5.5 mA with the standard-brightness LEDs JLCPCB stocks (v0.2.3); 1k5 for high-
    efficiency parts
LED12_R = 2200 # ohms: +12 V indicator series, ~4.5 mA
PULLUP_R = 10000
```